FMI	ADMINISTRATIVE INFORMATION	FMI No: 25339A

FMI Title: LightSpeed 2.X, 3.X, 4.X Performix Mounting Bolt Replacement

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
[X] Mandatory Safety	GEMS-AM:	Issue Date:
[ ] Mandatory	GEMS-E:	
[ ] Mandatory on Request	GEMS-A:	Close Date:

[X] **Staged FMI** — **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

[X] **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

[ ] **Parts Ordering**: GEMS - Americas, contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

[ ] **No Parts** are required to complete this FMI.

[X] **Batch with** — FMI 25335, FMI 13625 (Discovery ST)
To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).

[ ] **Non-Staged** — **No Customer List is provided**. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

"Preliminary Steps" (page 5) MUST be reviewed prior to any activity on this FMI.

Unit Tracked:

FMI Instructions Part No.: 2405344-100

Do NOT leave unsecured on site.

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
GE Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - E / ESC FMI Admin. 322 BP34 F 78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25339A **FMI Title**: LightSpeed 2.X, 3.X, 4.X Performix Mounting Bolt Replacement

Customer Full Name: ____________________

Number and Street Address: ____________________

City and Country: ____________________

____

Country Code **Site Number** **System I.D.**

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

1.) Modification Complete or previously modified.
2.) FMI not required per FMI instruction.
3.) Not modified. Unit in storage. FMI kit returned to stock.
4.) Modification refused or equipment altered. Customer signature and title required.
5.) Unit location unkown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer Comments ____________________

Effectivity

All of the following CT Systems, with the defined gantry types, are affected by this FMI:

- LightSpeed 2X (QX/I, Plus)
- LightSpeed 3X (QX/I, Plus, Ultra)
- LightSpeed 4X (16 Slice)
- HiSpeed QX/i
- Discovery LS (PET/CT)
- Discovery ST (PET/CT)

The following LightSpeed Family Gantry Model Types are affected by this FMI:

- 2260619
- 2260619-3
- 2281177
- 2281177-2
- 2281177-3
- 2281177-4
- 2305500
- 2339985
- 2341799
- 2362605
- 5101272

Purpose

The purpose of this FMI is to replace the four M10 Mounting Plate screws used to secure mounting plate 2128696 to Performix tubes for use on LightSpeed 2.x, 3.x, 4.x, HiSpeed QX/I, and Discovery LS/ST systems.

Bolt replacements are required per Product Safety Reports:

PSR 13014108, PSR 13015199 & PSR 13015422
(Tube Mounting Hardware Concerns)

There is a data sheet with this FMI that must be filled out and stored with the System log book. An online survey, identical to the data sheet, must also be filled out and submitted as part of this FMI.

If you are not a GE employee, you will not be able to submit online. Instead, you are required to e-mail the data sheet and return a paper copy via traditional mail. Directions are provided for this case.

NOTICE **You MUST review the safety presentation and training video information available on the supplied CD-ROM prior to performing this FMI. The material on this CD IS different from any previous FMIs.**

- **Torque Wrench Use for Tube Mounting (video)**
- **LightSpeed Family Performix Tube Change Safety (Power-Point - NEW)**

Prerequisite FMI

FMI Number	Title
FMI 25334	Performix Tube Field Inspection

Related FMI

FMI Number	Title
FMI 25335	Installed Base Performix Tube Detailed Evaluation

Furnished Materials

The following parts are provided with this FMI

Qty	Description	Part #
1	FMI Document	2405344-100
1	FMI 25339 Service CD-ROM	2405439
1	LightSpeed Tube Bolt Kit	2391264-2
1	Tube Bolt Measuring Block	2405441
1	FMI 25339/25340 Complete Label	2405440
1	LightSpeed General System Service Manuals CD	2405438
1	Red Defective Sticker	2358146

Required Tools

The following hardware tools are required to perform this FMI:

Qty	Description	Part #
1	Torque Wrench Kit	*Per GEMS Service Regional Team* 46-268445G1
1	Tube Hoist	*Per GEMS Service Regional Team* Arm: 2282509, Boom: 2282510 (Americas)

The following Service Documentation CD-ROMs are needed to complete this FMI. The FMI instruction will identify when these documents are to be used

Qty	Description	Part #
1	LightSpeed Series Service Information CD-ROM	2360027-200, Rev 04
1	Discovery Series Service Information CD-ROM	2371120

People Power

8 hours labor; 1 hour travel

Before You Begin

1.) If you installed a new tube on the system and it has one of the following part numbers, skip to FMI Completion, on page 14.
- D3185T (Mounting Plate attached, see Figure 1) or 5110958
- D3186T ((Mounting Plate attached, see Figure 1) or 5110958-2

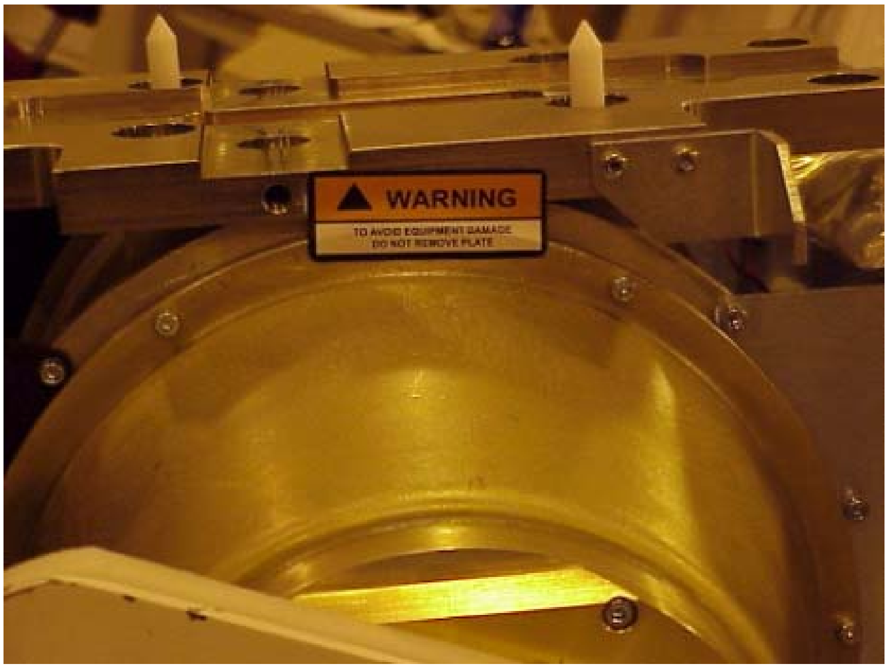


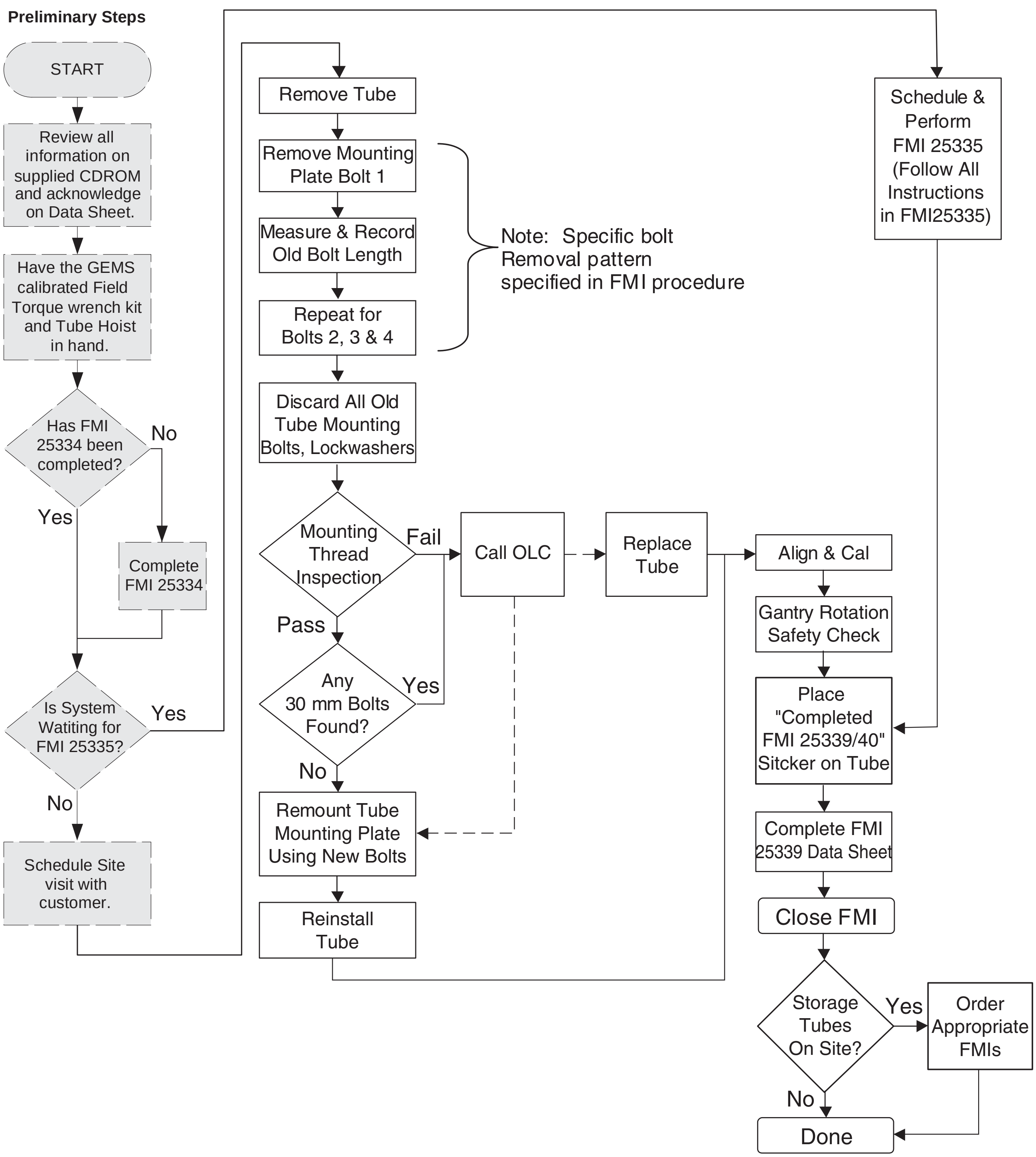
Figure 1 - Tube with mounting plate attached

2.) If the gantry manufacture date is August, 2003 or newer and the tube has NOT been changed or removed, skip to FMI Completion, on page 14. You need to add a note in your dispatch stating “Gantry manufacture date is August 2003 or newer”. This is for MKE and Hino made gantries, including PET systems. Record this on the data sheet.
- If you replaced the tube and the tube is one of the following part numbers you can skip to FMI Completion, on page 14.
  * D3185T (Mounting Plate attached, see Figure 1) or 5110958
  * D3186T ((Mounting Plate attached, see Figure 1) or 5110958-2
- If you have removed the tube for any reason you need to go to Preliminary Steps and complete the FMI.

Procedure

NOTICE **Make certain to read the information in the section titled "Before You Begin," on page 3, before performing any of the steps in this FMI.**

1 Performix Mounting Plate Bolt Replacement Process Flow



2 Preliminary Steps

1.) Review all information on the supplied CD-ROM (P/N 2405439) prior to going to site.

2.) There is a data sheet on page 16 of this FMI document that must be filled out and stored with the system log book. It is **strongly recommended** that you copy the data sheet and keep one for your records. An electronic copy of the data sheet is supplied on the FMI CD-ROM (2405439).

3.) There is a web survey found at http://www.ct.med.ge.com/tech_pubs/FMI25339datasheet.htm that must be filled out prior to closing this FMI. For non-GE employees, alternate submittal instructions are provided with the data sheet.

4.) If a system is waiting for FMI25335, go to FMI Completion, on page 14.

3 Mounting Bolt Replacement

CAUTION **Tube and tube oil may be hot, depending on recent scan load. Wait 30 minutes after last scan, with power to the gantry, to allow the tube and tube oil to cool, prior to removal.**

Note: Some Performix Tubes were inspected during prior FMIs and identified as “4-weld suspect tubes.” You may see a 4-weld tube that has a sticker on it from manufacturing (see Figure 2, below). Such tubes are OK for use. Continue performing the FMI.

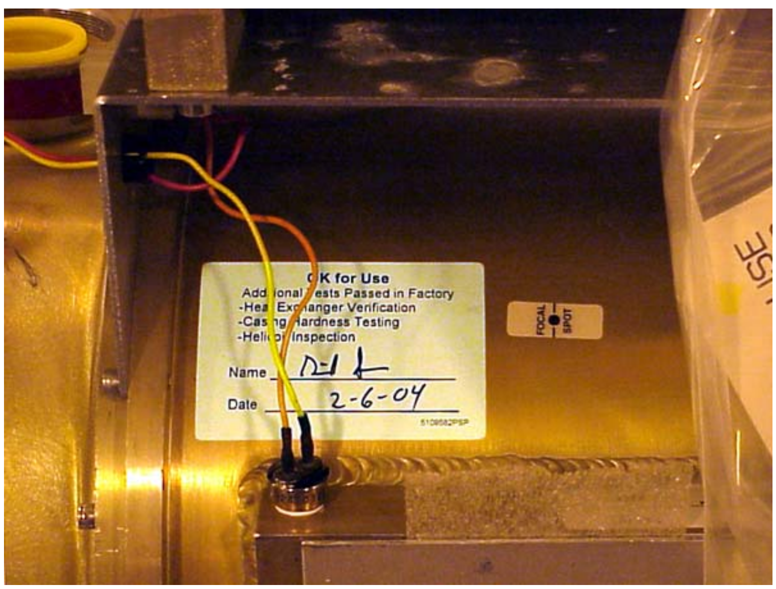
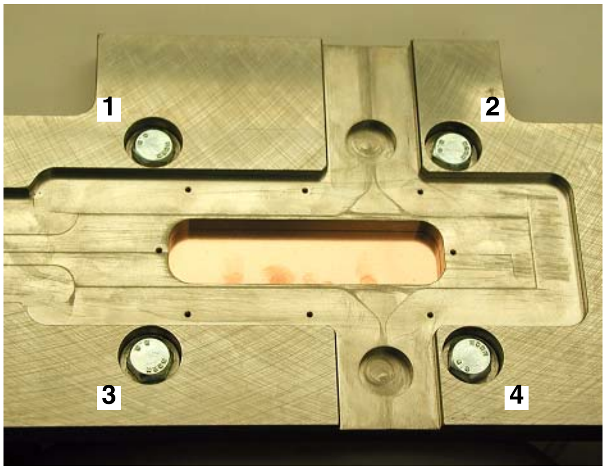


Figure 2 - “OK for Use” Sticker

1.) Use standard service procedures specific to your system type for removing the tube, but DO NOT remove the mounting plate (until instructed to, in the next step). Refer to "Remove Old Tube" in Appendix A - Tube Replacement: HiSpeed QX/i, LightSpeed, Discovery LS/ST.

2.) Remove Bolt 1, as identified in Figure 3, below.



Heat Exchanger
On This Side

Figure 3 - Tube Mounting Plate Bolt Locations

3.) Using the measuring block provided, verify that the bolt is a M10 x 25mm bolt. Make certain to fit bolt *inside* the block, *not outside* (see Figure 4*).* M10 x 25 mm bolt on right fits completely within measuring block. Note that the 30 mm long bolt does not fit in the block with the bolt head in the slot.

If the bolt does NOT fit in the block, then it FAILS the test.

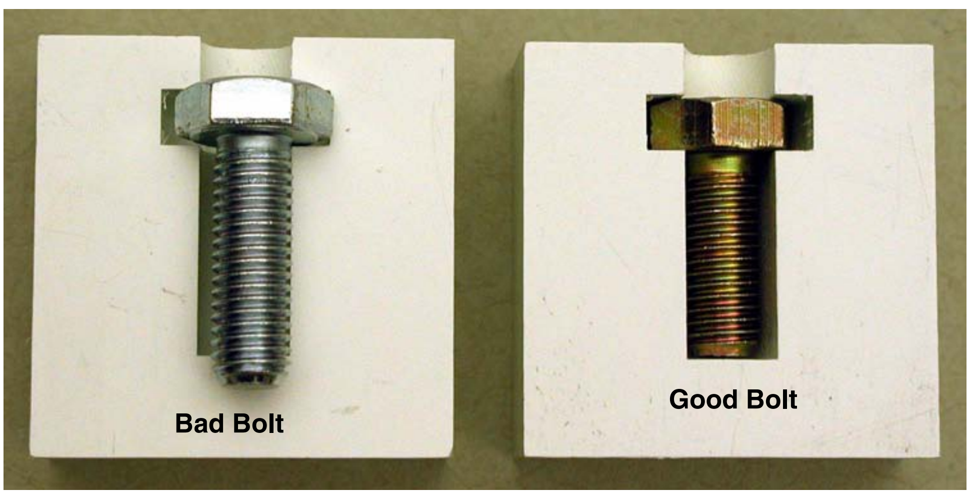


Figure 4 - Measuring Block Usage: 30mm bolt shown at left, 25mm bolt shown at right

4.) Record the result on the data sheet on page 16.

5.) Removing bolts one at a time, repeat the same procedure (steps 2-4) for Bolts 2, 3, and 4.

6.) Do NOT reuse any of these removed bolts. Discard them now.

7.) Remove mounting plate, for mounting hole inspection.

4 Mounting Hole Inspection

Perform a visual inspection of the mounting holes in the tube case, using the process defined in this section. You are looking for any types of unsafe conditions, primarily physical damage to the helicoil or underlying base threads in the aluminum. Some examples are provided as a guide, but defects are not limited to these cases. You may want to use a flashlight for good lighting, and a magnifying glass.

A helicoil is a threaded insert used in a threaded mounting hole, for increased mounting strength. The helicoils used in the four mounting holes joining the tube and collimator plate are "locking helicoils." This means that they have a squared section in the middle, as shown in Figure 5, below. Do not consider the squared section to be a thread defect.

If you have any questions, call the Online Center (OLC) (for Americas: Select "Quick Question").

If any mounting holes look suspect, you must call the OLC (for Americas: Select "Quick Question").

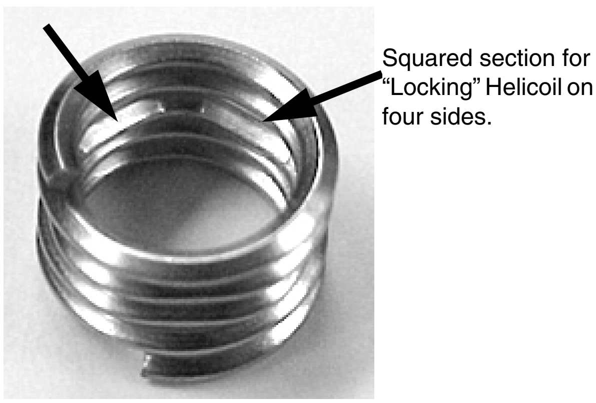


Figure 5 - Helicoil Insert Outside Tube

1.) Using the pictures above and below, make sure the helicoil is recessed properly below the surface of the tube casing plate. There should be a full thread above the beginning of the helicoil. Verify the locking thread, as shown above, is present on each helicoil. If the locking thread is not present, the tube should be replaced.

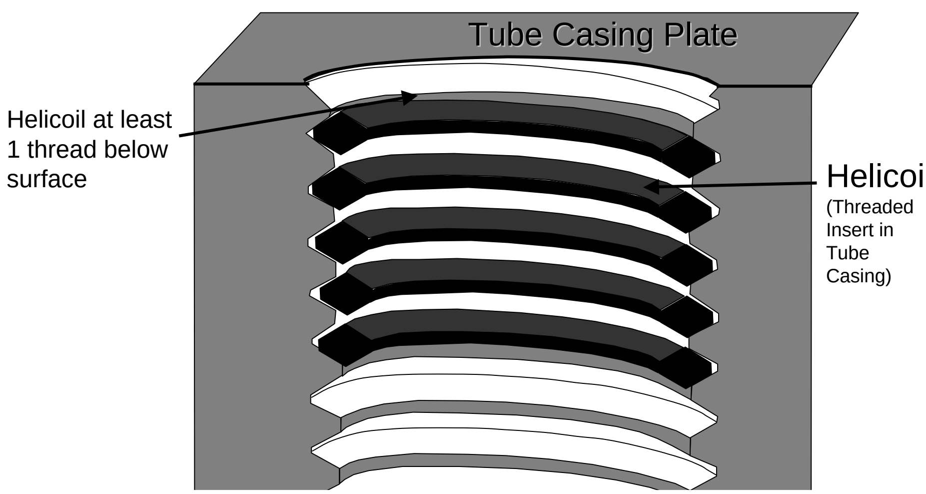


Figure 6 - Cross-section of tapped hole with Helicoil Insert

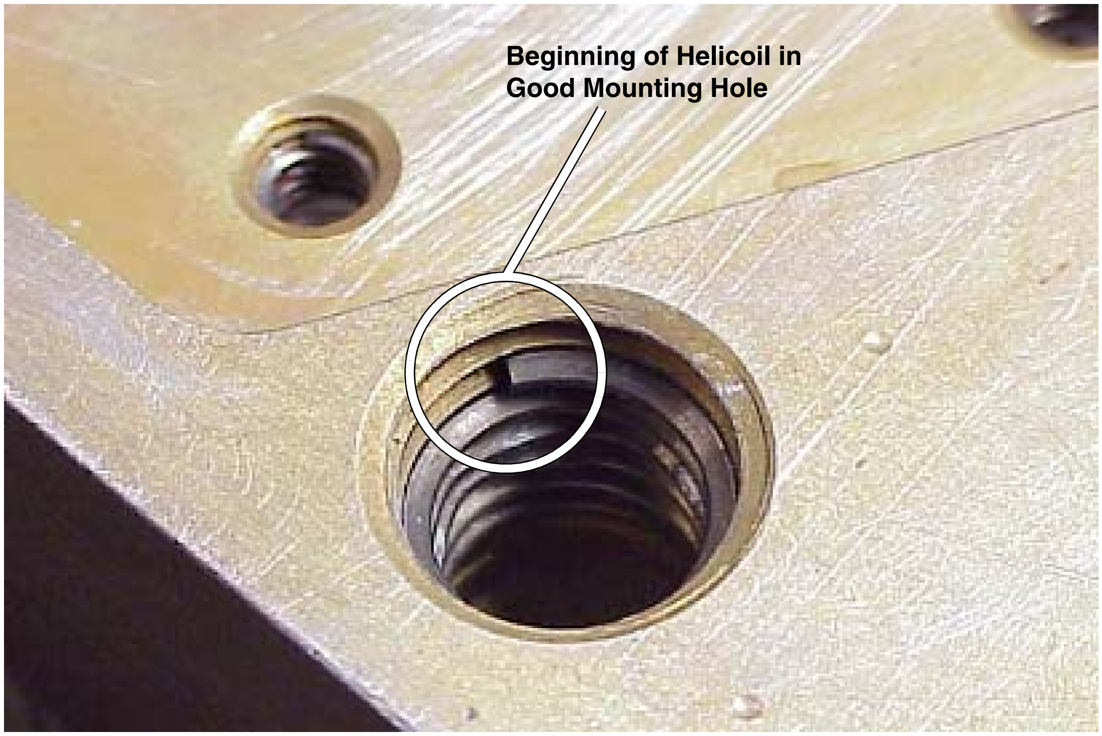


Figure 7 - Beginning of Helicoil

2.) Using compressed air, blow out hole to remove loose particulate. Remember to wear safety glasses while using compressed air. Visually inspect helicoil. Trace helicoil through all threads, looking for continuity and smoothness. There should be no breaks in the helicoil. Inspect for signs of gouging, tearing, or any gross material removal of the helicoil. If ANY of these are present, you must call the OLC (for Americas: Select “Quick Question”) for further instruction. They will assist in validating that the tube needs replacing.

Note: The squared off locking thread as shown in Figure 5 is normal and should be visible.

3.) Visually inspect for any damage to threads. Examples include, but are not limited to those shown in the following figures.

Note: Figures may be best viewed on a PC, for color.

Figure 8 - Helicoil pulling out of casing

Figure 9 - Large particulate jammed in thread

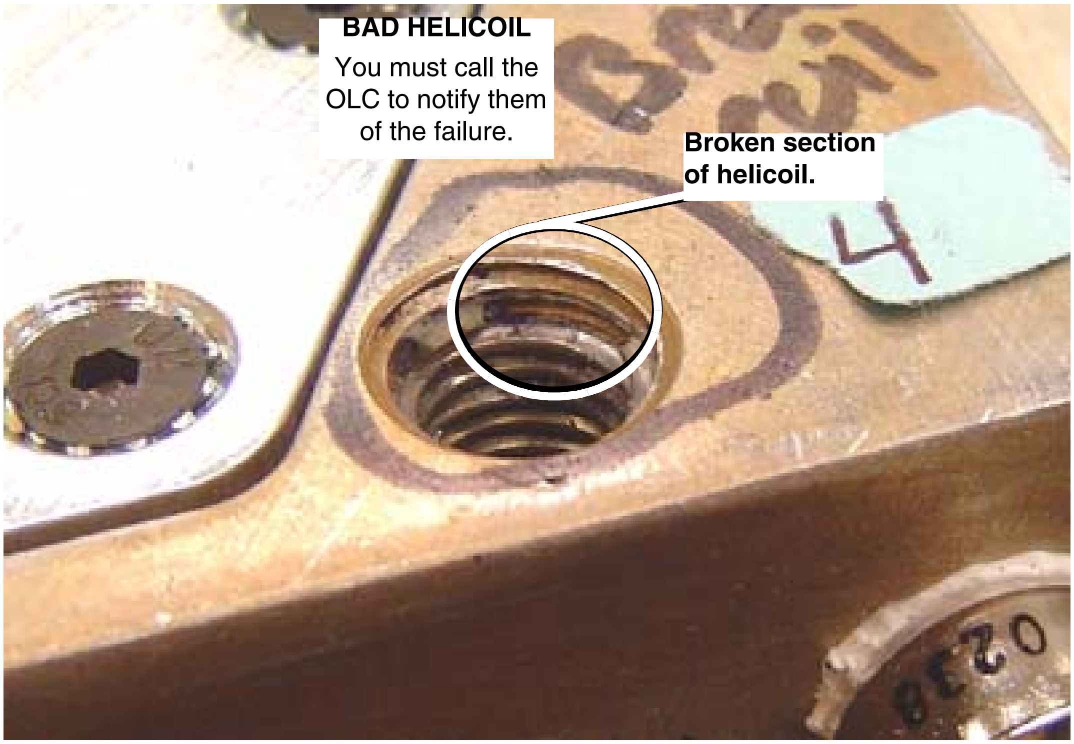


Figure 10 - Broken Helicoil

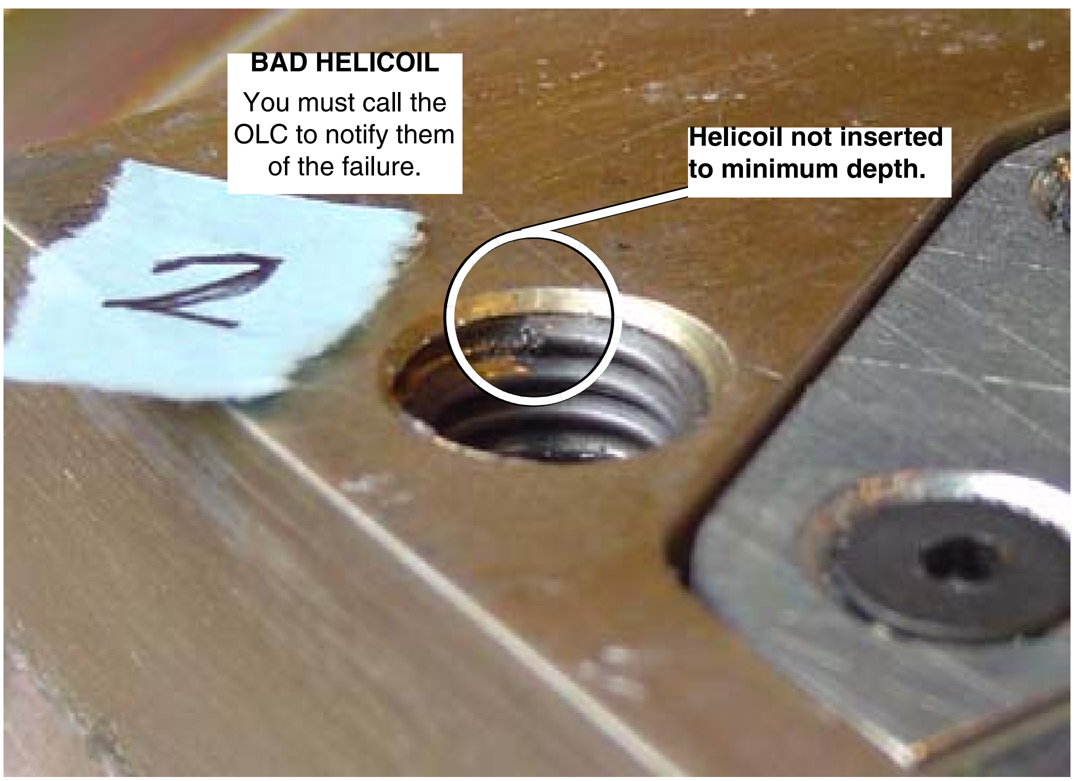


Figure 11 - Helicoil not inserted to minimum depth

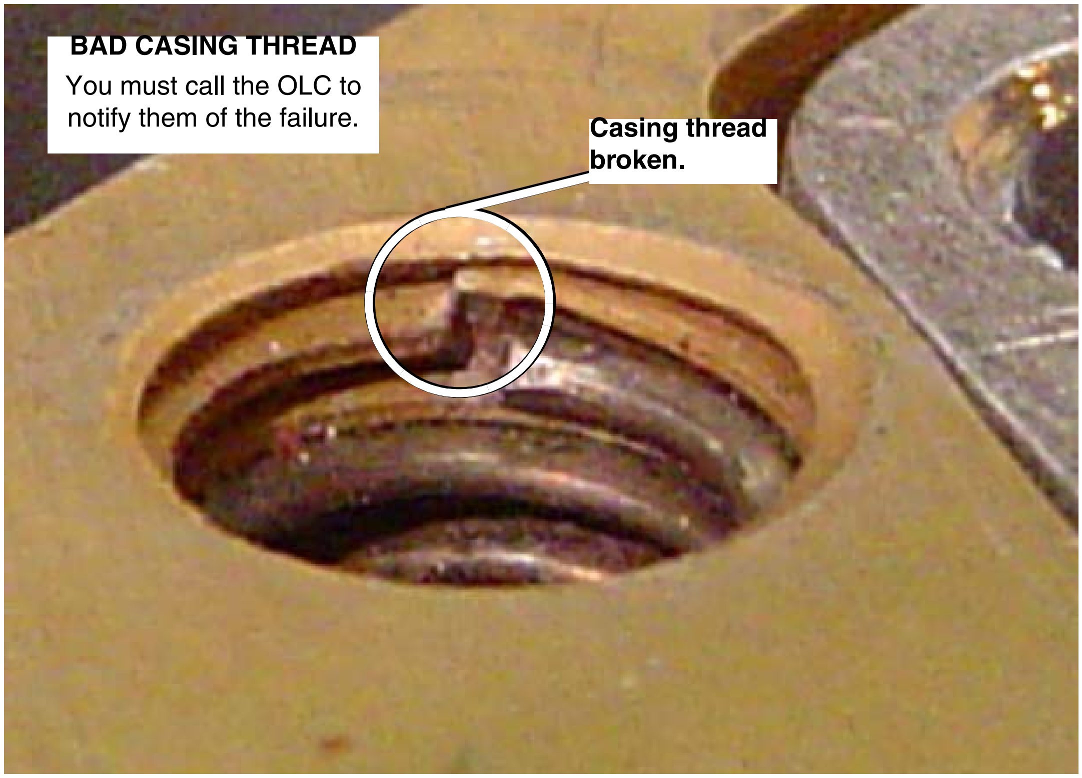


Figure 12 - Casing Thread Broken

- Record inspection results on data sheet on page 16.
- If ANY mounting hole fails the inspection, you must call the OLC (for Americas: Select "Quick Question") for further instruction. They will assist in validating that the tube needs replacing.
- If all mounting holes are visually OK, AND:
  - **All bolts fit** in the measuring block, skip to Tube Installation, on page 12.
  - **All bolts DID NOT fit** in the measuring block, call the Online Center (for Americas: Select "Quick Question")

5 Tube Replacement

5.1 Order New Tube

Replace the tube, if any thread inspection fails. You will need to inform the customer that the tube has to be replaced, along with estimated time for tube arrival. Also inform the local sales and service managers that the customer will be down for a tube change.

QTY	PART #	DESCRIPTION	USED ON
1	D3186T	Performix Ultra Tube:	LightSpeed 2X, 3X, 4X, and Discovery LS, ST

NOTICE **If you are on a government site that requires new replacement parts instead of refurbished parts use the following table to order the proper tube.**

QTY	PART #	DESCRIPTION	USED ON
1	D3185T	Performix Ultra Tube:	LightSpeed 2X, 3X, 4X, and Discovery LS, ST

5.2 Tube Installation

WARNING **ONLY USE THE TORQUE WRENCH TO PUT IN TUBE BOLTS. THIS IS A LOW TORQUE BOLT. USE OF ANYTHING OTHER THAN A TORQUE WRENCH MAY RESULT IN THREAD DAMAGE.**

NOTICE **REQUIRED: View the video titled “Proper use of a Torque wrench” and the “LightSpeed Family Performix Tube Change Safety” slides on the CD-ROM (P/N 2405439) supplied with this FMI, prior to installing the tube mounting plate and tube.**

1.) Use NEW Bolt Kits (P/N 2391264-2), provided with the replacement tube, to attach:
   a.) Mounting Plate to the Tube
   b.) Tube to the Gantry

Note: The four M10x25mm bolts are gold colored.

CAUTION **ONLY use the bolt kit labeled for the system upon which the tube is used.**

2.) Install the tube per instructions provided in Appendix A - Tube Replacement: HiSpeed QX/i, LightSpeed, Discovery LS/ST.

NOTICE **If reinstalling the same tube, the tube must cool for five (5) hours after the last scan is run on that tube.**

3.) Align and calibrate your system per standard service procedures for your system type.

5.3 Returning Tube

Attach one of the supplied Red “Defective Tube” sticker on the tube, in the location shown in Figure 13, below. Place the other defective tube sticker on the tube crate. Return the tube within 24 hours using standard tube return process. Be sure to record the Waybill number and shipper name.

Figure 13 - “Tube Defective” Sticker Location

FMI Completion

1 Instructions

- Four (4) tube mounting plate bolts replaced, using Performix Mounting kit for Performix Systems (P/N 2391264) and four M10x25mm gold colored bolts for attaching mounting plate to tube. FMI 25335 and 25339 both instruct this to be done.

  If the tube was replaced by one of the new part number or the gantry manufacture date is August 2003 or newer and the tube was not removed or changed, you don't need to complete this step. The new parts numbers are:
  - D3185T (Mounting Plate attached) or 5110958
  - D3186T (Mounting Plate attached) or 5110958-2
- Place “Completed FMI 25339/40” sticker on tube.
- Deliver Tech Pubs *LightSpeed Family System Service Manuals* CD-ROM (2405438) to customer. Recommend that the customer discard any other service manuals on site.
- Data sheet (page 16) filled out and submitted:
  - GE Employees: Web data sheet completed, submitted and confirmed
  - Non-GE Employees: Paper and e-mail data sheet sent
- Verify customer has no X-Ray tubes in storage. If customer DOES have tubes in storage, order FMIs per Table 1, below.

Note: If FMI 25335 (Installed Base Performix Tube Detailed Inspection) has already been completed on this system, and no tube change has occurred since the closure of FMI 25335, you may close this FMI. However, you must verify that the tube identified during FMI 25335 is still on the gantry and place the “Completed FMI 25339/40” sticker on the tube. You must also submit the FMI 25339 data sheet before closing this FMI.

2 If Customer Has Tubes In Storage

Table 1, below, lists the FMIs to order according to which tubes the customer has in storage.

Table 1 - Customer Tubes in Storage

X-Ray Tube in Storage	FMI(s) to Order
D3172T D3171T	FMI 25340 FMI 25340
D3182T D3181T	FMI 25334 & FMI 25339 FMI 25334 & FMI 25339
D3175T (Mounting plate attached) or 5110958-3 D3176T (Mounting plate attached) or 5110958-4	FMI 25340 - Put completion label on the tube FMI 25340 - Put completion label on the tube
D3185T (Mounting plate attached) or 5110958 D3186T (Mounting plate attached) or 5110958-2	FMI 25339 - Put completion label on the tube FMI 25339 - Put completion label on the tube

3 Additional FMI Completion Instructions

You MUST follow all applicable Instructions before closing this FMI.

1.) Submit the Data Sheet

If you are a GE Employee:

Visit http://www.ct.med.ge.com/tech_pubs/FMI25339datasheet.htm and submit your data electronically. This will take you five minutes, once you are connected. You should enter your e-mail address, so that you receive a copy of the data sheet that you submit. When you submit, a confirmation screen will appear.

If you are NOT a GE Employee:

Submit the data sheet found on the CD-ROM (2405439) via e-mail, to: ***performix.inspection@med.ge.com***. The e-mail subject should read “FMI 25339".

AND

Mail a signed copy within three (3) days to:

Andrew Rubin, EA36
GE Healthcare - Tubes
4855 W. Electric Avenue
Milwaukee, WI 53219

2.) Update Global Installed Base (GIB) database on the web. Remove the tube casing serial numbers and tube insert serial numbers that are no longer on this CT System. Go to Appendix B - Global Installed Base (GIB) Update Instructions, and follow the process specific to your region.

3.) If a tube is replaced:

Load the Product Locator Card information into GIB (see step 2, above).

4.) Remove the FMI Document from the customer site.

FMI 25339 Data Sheet

PERFORMIX INSPECTION CHECKLIST

Complete this Data sheet. Leave a copy with the System and keep the original for your records. Directions are provided on the previous pages.

All procedures may not have been done. Document only those inspections that were actually performed.

Initial the boxes below to confirm that you have reviewed the supplied videos prior to performing this FMI. Review all information on the supplied CD-ROM (P/N 2405339).

	Torque Wrench Video	Performix Tube Change Safety Slides
CD-ROM Materials Reviewed		

yes / no	Did you perform FMI 25335 (Installed Base Performix Tube Detailed evaluation) on this site?
yes / no	Gantry August, 2003 or newer?

Initial boxes to indicate you have performed the following:

- [ ] Old mounting plate bolts discarded.
- [ ] New mounting plate bolts used to mount plate and tube.
- [ ] Mounting hole inspection.
- [ ] “LightSpeed Family System Service Manuals” CD given to customer.

Complete the tables below for the system being inspected:

Torque Wrench Tracking Number	

Hospital Name	
GEMS System ID	
System Type (Circle)	LightSpeed 2.x, 3.x, 4.x, HiSpeed QX/i, Discovery LS/ST
Gantry Model Number	
Gantry Serial Number	
Insert Serial Number	
Casing Serial Number	
Date of Manufacture	
Date of Last Tube Install	

FE Name		Date FMI Executed	

	Location/Hole 1	Location/Hole 2	Location/Hole 3	Location/Hole 4
Mounting Plate Bolt Measurement	Fit / No Fit	Fit / No Fit	Fit / No Fit	Fit / No Fit
Mounting Hole Inspection	Pass / Fail	Pass / Fail	Pass / Fail	Pass / Fail

Tube Returned?	Yes / No
Waybill Number:	

Reason (circle):	Failed Thread Inspection / Other
Shipper:	

Appendix A - Tube Replacement: HiSpeed QX/i, LightSpeed, Discovery LS/ST

1 Introduction

1.1 Purpose

This document covers the quick replacement procedure for X-Ray Tubes. See Figure 14 for an overview of this process. The sections following explain this process.

1.2 Requirements

The system must meet the following requirements to perform the Quick Tube Replacement Procedure:

1.) Normal Tube Change with no other system issues.

2.) The previous Tube was aligned correctly. (POR, BOW, CBF, ISO were in spec.)

3.) The Collimator and the Detector have not moved.

4.) There are no IQ issues.

5.) If kV gain pots are adjusted, you must run a full set of calibrations.

If these requirements are met, the Quick Tube Replacement Procedure can be followed. If the system does not meet these requirements, the full Tube alignments and calibrations described in the appropriate *System Service Manual* must be performed.

1.3 Quick Tube Replacement Process Flow - Same Tube Re-Installed

Follow this flowchart, if you remove the tube from the gantry and re-install the same tube back onto the gantry.

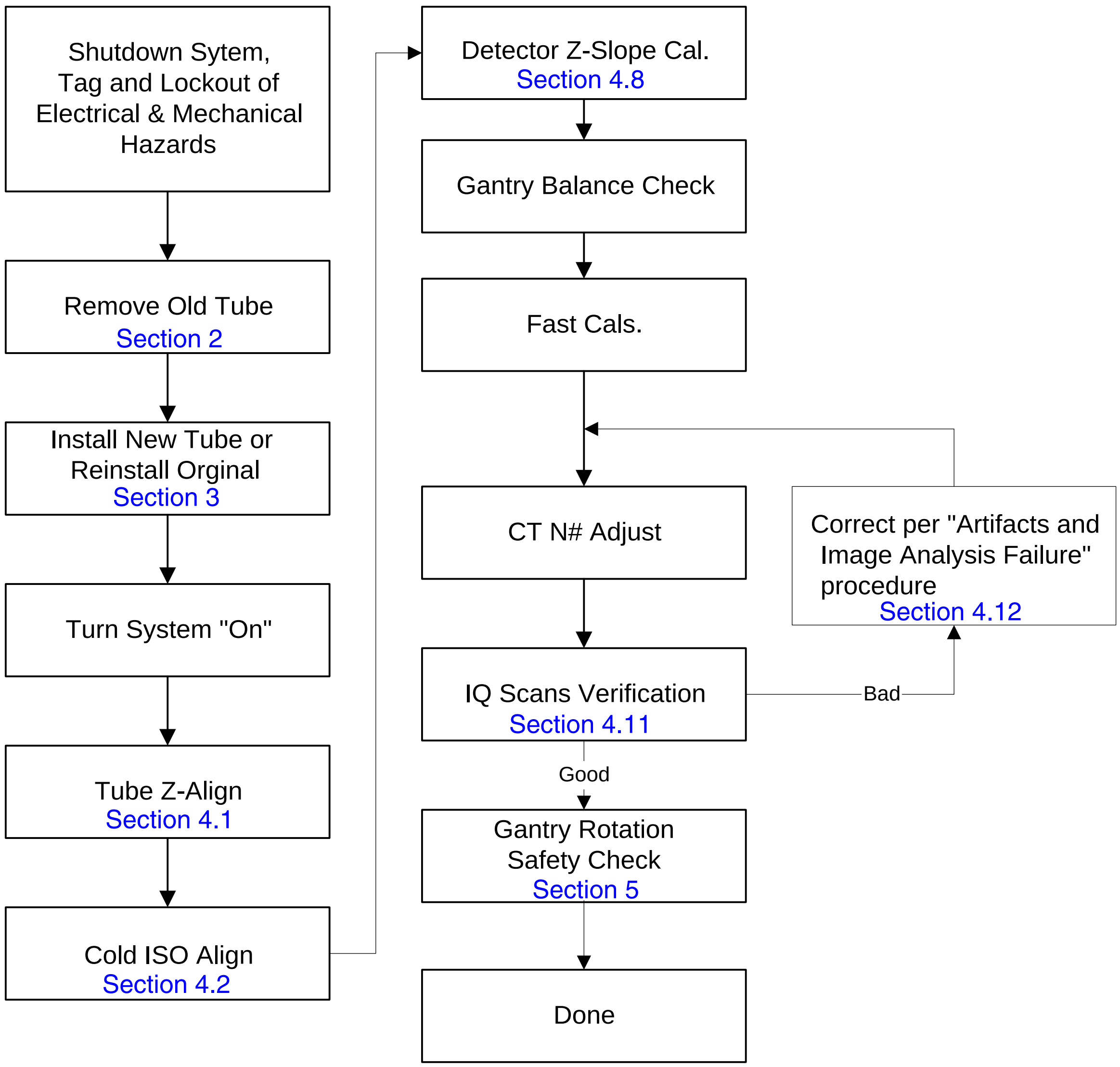


Figure 14 - Quick Tube Change Flow

1.4 Quick Tube Replacement Process Flow - Tube Replaced

Follow this procedure, if you replace the tube.

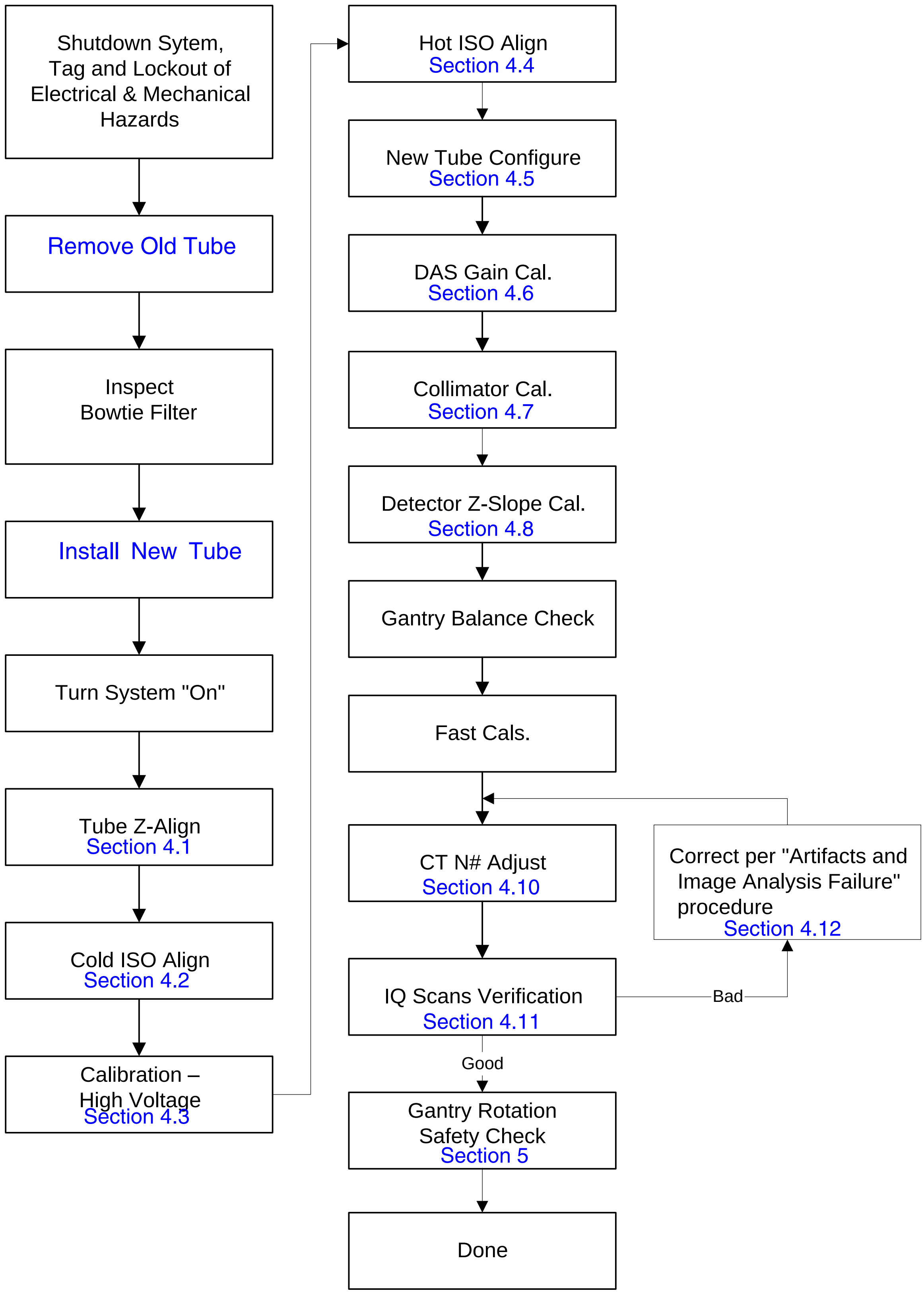


Figure 15 - Quick Tube Change Flow

2 Remove Old Tube

NOTICE **If the tube has a label as shown on Figure 1, on page 3, DO NOT remove the tube. Skip to FMI Completion, on page 14.**

Before beginning this procedure, please read the gantry safety information.

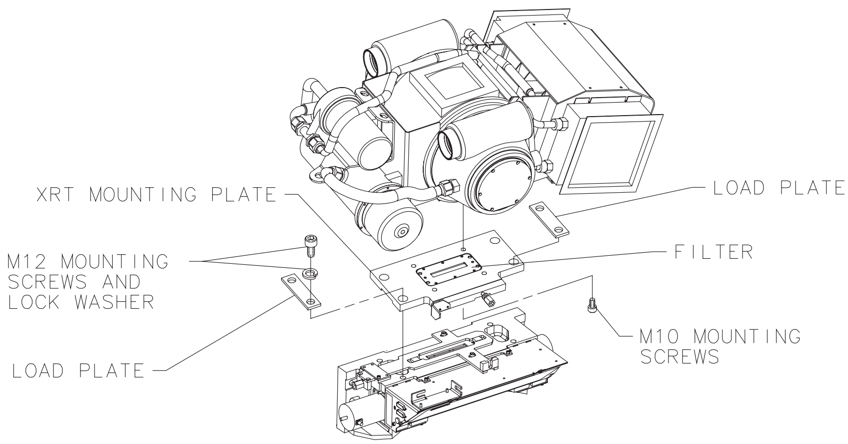


Figure 16 - Tube Removal Diagram

WARNING **FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.**

Make sure you engage the locking mechanism before you remove the tube. Failure to lock the gantry could result in injury, should the gantry suddenly move and strike you.

1.) Remove, and set aside, both gantry side covers, top covers, front and rear covers.
2.) Remove the M12 screws from the right front gantry cover mounting bracket, remove and set aside the bracket. Reference Figure 17.

NOTICE **It might be necessary to tilt the gantry back to remove the third bolt, which is not normally installed. Remember to tilt the gantry back to zero degrees.**

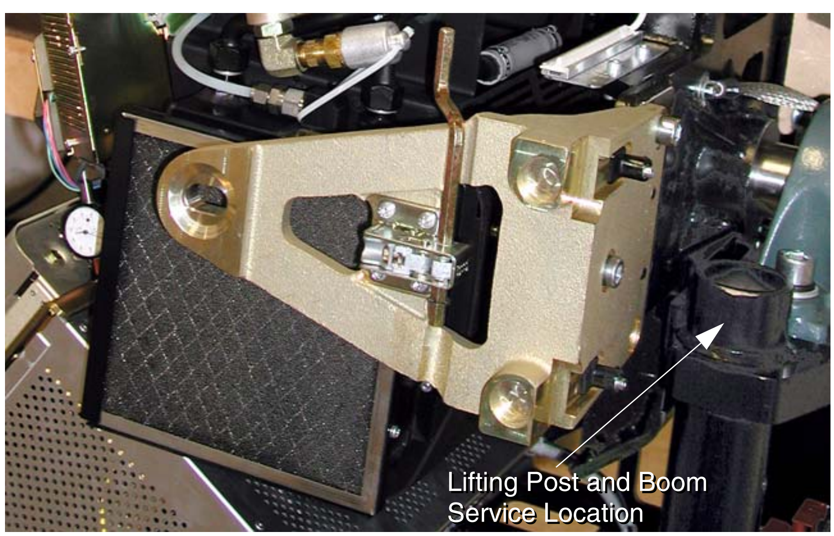


Figure 17 - Right Front Gantry Cover Mounting Bracket

3.) Turn off facility power to **PDU** and lockout/tagout.

4.) Turn off the **AXIAL DRIVE ENABLE** and **HVDC ENABLE** switches on the **STC** backplane.

NOTICE **It may be easier to loosen the M12 tube mounting bolts with the tube at about the 2 o'clock position before locking the tube at the 3 o'clock position. Simply loosen the tube mounting bolts one half (1/2) turn. Do not remove the bolts yet.**

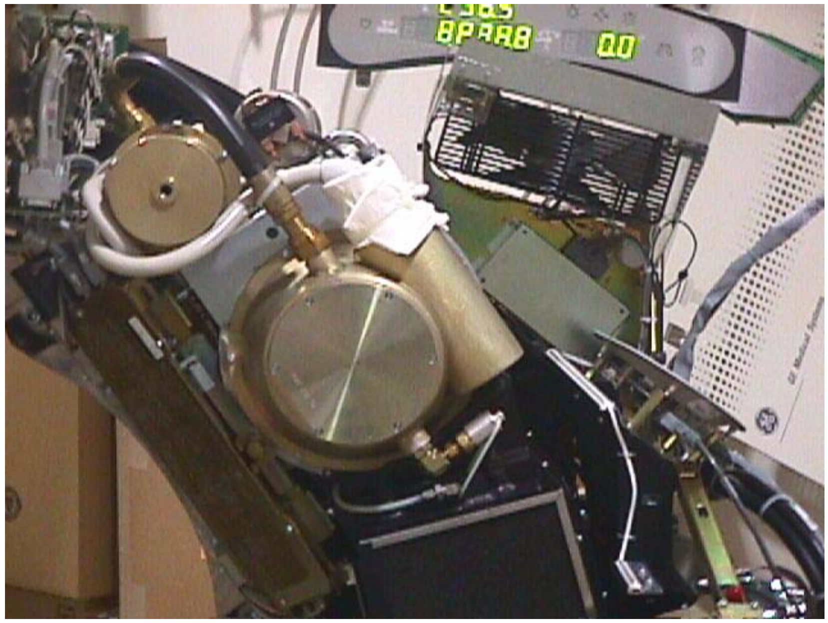


Figure 18 - Tube Angle to Loosen and Torque M12 Screws

5.) Rotate the Gantry until the failed tube unit reaches the 3 o'clock position.

NOTICE **Make sure the tube is at 90 degrees so the tube hangs at the correct engagement angle for removal and installation.**

6.) Engage rotational lock. Check that the gantry is securely locked by attempting to rotate the gantry by hand.

7.) Insert the lifting post, boom and chain hoist. Reference Figure 17.

8.) Disconnect the 12 pin tube I.D. system cable, from the top of the tube unit.

9.) Disconnect the 4 pin mate-n-lock pump and fan power system cable

10.) Disconnect the ground strap from the top of the tube unit

11.) Remove the anode and the cathode cable:

- Carefully cut ty-raps securing **HV** cables. Note **HV** cable routing.
- Loosen each cable's locking ring with the spanner wrench.
- Pull each cable terminal out of its receptacle.
- Ground the end of the cables to the Gantry frame.
- Wipe up any oil that drips from the cable terminal.
- Use paper towels to soak up any oil in the wells.

CAUTION **Remove the mounting bars in the following (lower/upper) order to lessen the risk of injury to your hand. Keep one hand under the bolt and pressure plate while unfastening it. This is to prevent them from falling into the fan that is attached to the tube.**

Note: It may be easier to tape the socket to the extension. This will prevent the socket from being dislodged on the tube radiator assembly

12.) The XRT is attached to the Collimator with a Tube Mount Bracket Assembly (P/N 2128696). Remove the mounting plate & XRT from the Collimator by removing the four M12 (P/N 46-328416P24) cap screws and lock washers, and two load plates (P/N 2120104), with a hex socket driver. With your hand, reach behind the radiator to the screws from either side of the XRT center section while removing the bolts with two 12 inch extensions (24 inch length) on a ratchet. **Throw these M12 bolts and washers away, as they should not be reused.**

13.) Carefully swing the tube clear of the gantry.

Note: **Be careful not to damage the fragile copper filter or lead shield in the mounting plate for the next step.**

CAUTION Potential for IQ artifacts **When removing the mounting plate from the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image by affecting the attenuation properties of the copper filter.**

14.) If the new tube has a factory installed mounting plate (see Figure 19), **DO NOT REMOVE IT**. Skip steps 15 and 16.

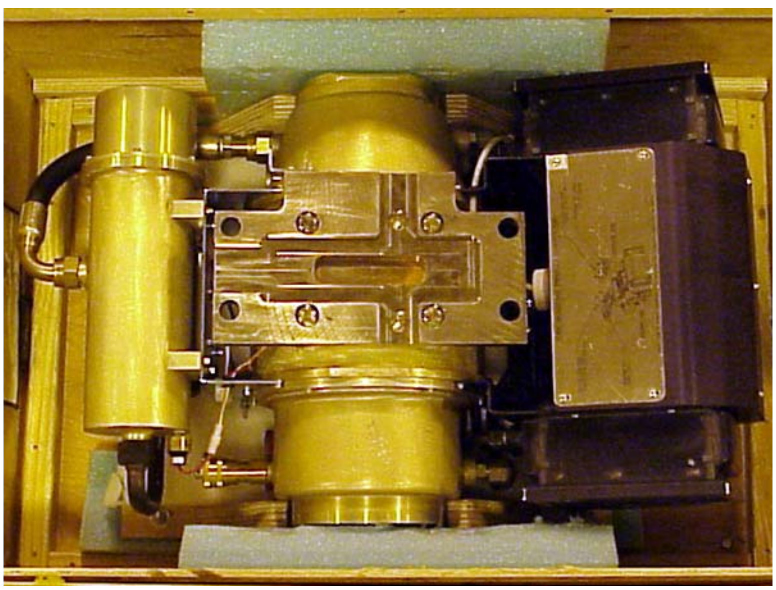


Figure 19 - Tube Shipped with Mounting Plate Attached

15.) If the replacement tube does NOT have a mounting plate attached, remove the mounting plate from the old tube by removing the four M10 (P/N 46-328416P20) hex head screws. **Throw these bolts and washers away, as they should not be reused.**

16.) If the replacement tube does NOT have a mounting plate attached, inspect the copper filter. The copper filter should be clean, dent, and scratch free:

- If contamination is visible (see Figure 20), proceed to the cleaning process, as found in the appropriate *System Service Manual*.
- Discoloration is acceptable.

Figure 20 - Extremely Contaminated Copper Filter

17.) Inspect the Bowtie Filter. It is possible that the Bowtie Filter is contaminated and the Primary Copper Filter is not contaminated.

Note: Perform this inspection before installing the new tube unit.

The following tools are required for this inspection procedure:

- Phillips #2 Screwdriver
- Flat blade Screwdriver
- Bright Flashlight

Inspect the Bowtie Filter as follows:

a.) Remove the Collimator Control Board Sheet Metal Cover. See Figure 21.

Figure 21 - Collimator Control Board Cover

WARNING **DO NOT ATTEMPT TO MOVE THE FILTER WITH POWER ON.**

b.) With power removed from the gantry, use a flat blade screwdriver and position the bowtie filter assembly so it is visible through the input port or tube side of the collimator assembly.

* CCW will move the filter into the beam. See Figure 22.
* Do not move the filter back to the home position.

NOTICE Potential for equipment damage

Do not force the filter if it feels stuck. Damage to the limit switch can result.

Do not move the filter more than necessary for inspection. The filter can fall off the drive screw.

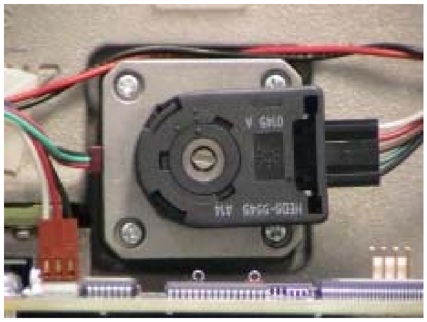


Figure 22 - Filter Position Adjuster

c.) Using a flashlight, inspect the bowtie filter for contamination.
Look through the input port or tube side of the collimator.
If contamination is visible, proceed to cleaning process.

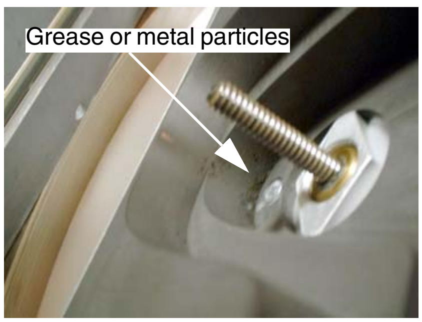


Figure 23 - Contaminated Bowtie Filter

3 Install New Tube or Re-Installing Original Tube

WARNING **SEVERE INJURY POSSIBLE.**
TUBE CAN SEPARATE FROM GANTRY, IF NOT TORQUED CORRECTLY.
ALWAYS USE PROPER TORQUE ON ALL FASTENERS.

WARNING **FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.**

1.) Allow the tube unit to warm to room temperature before you install it.

2.) If the new tube has a factory installed mounting plate (see Figure 19 and Figure 1), **DO NOT REMOVE IT**. Skip to step 4.

WARNING **SEVERE INJURY TO PATIENT COULD OCCUR.**
USING THE WRONG BOLTS COULD PLACE STRESS ON BOLTS, CAUSING TUBE TO SEPARATE FROM GANTRY.
USE PROPER BOLTS.

3.) Attach the mounting plate from the old tube using the **four new M10 bolts** that come with the tube. **Do NOT use Loctite.**

- Finger tighten all four (4) bolts
- Torque all four (4) bolts to the following pre-load torque specification:

15 lb-ft	**20 N-m**	**180 in-lbs**	**210 kg-cm**

This seats each bolt, enabling you to visually ensure that the mounting holes are not stripped while applying final torque.

WARNING **SEVERE INJURY POSSIBLE.**
IF YOU TURN THE TORQUE WRENCH MORE THAN 90º (¼ TURN) WHILE APPLYING FINAL TORQUE, THE MOUNTING THREADS ARE STRIPPED.
DO NOT USE THIS TUBE.

- Set final torque on all four bolts:

30 lb-ft	**41 N-m**	**360 in-lbs**	**420 kg-cm**

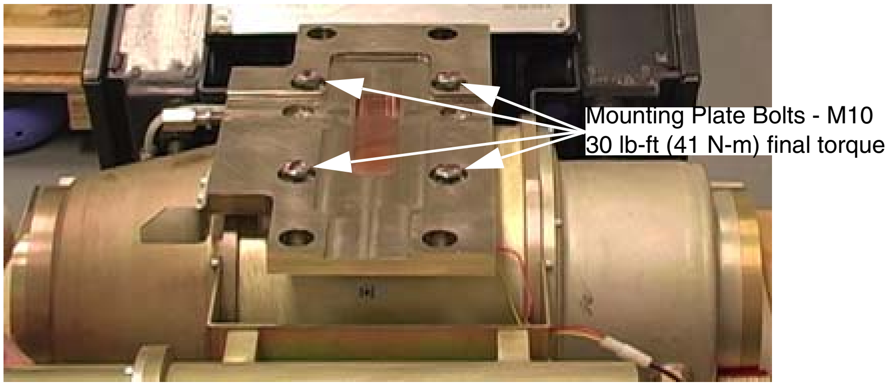


Figure 24 - Tube and Mounting Plate

CAUTION Potential for IQ artifacts

When attaching the mounting plate on the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image. Use a lint free wipe to clean, if needed.

4.) Re-check facility power and make sure it is off.

WARNING

SEVERE INJURY TO PATIENT COULD OCCUR.

USING THE WRONG BOLTS COULD PLACE STRESS ON BOLTS, CAUSING TUBE TO SEPARATE FROM GANTRY.

USE PROPER BOLTS, PER SYSTEM TYPE.

5.) Use the hoist to lift the new tube unit:

- Position the tube on the gantry: The “crosses” on the mounting plate and on the collimator should fit in perfectly when the tube is aligned properly.

Note:
  * To ease installation, fasten the top pressure plate to the rotating structure first. Then attach the bottom pressure plate.
  * **Use new bolts and washers from tube crate. Make sure to select the proper bolts. There are instructions in the crate, and on the tube itself.**

- Fasten the lower and upper and pressure plates to the rotating structure with the four M12(50 mm) bolts, and set **pre-load torque to:**

25 lb-ft	**33 N-m**	**295 in-lbs**	**340 kg-cm**

- Set final torque to:

49 lb-ft	**66 N-m**	**590 in-lbs**	**680 kg-cm**

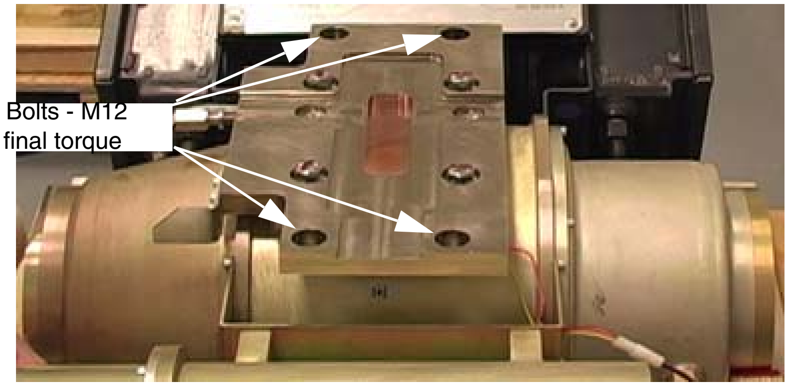


Figure 25 - Tube and Mounting Plate

6.) Attach the tube I.D. cable to the 12 pin mate–N–Lock connector on top of the tube.
7.) Attach the tube pump and fan power cable to the 4 pin mate–N–Lock connector.
8.) Fasten the grounding strap to the 1/4–20 ground stud on top of the tube unit.
9.) Remove the plastic cap plug from each cable receptacle on the tube unit.

Note: Take care not to lose the rubber quad rings for the High Voltage cables.

10.) Lightly wet the new rubber quad ring with transformer oil (917).
11.) Return the quad ring to its slot at the top of the receptacle retaining ring.
12.) Pour transformer oil (917) into the receptacle to a depth of 10 mm (0.375 in).

NOTICE **Incorrectly routed or secured HV cables will result in damage to the HV cables and/or other parts of the gantry.**

13.) Be sure to route the **HV** cable as shown in Figure 26.

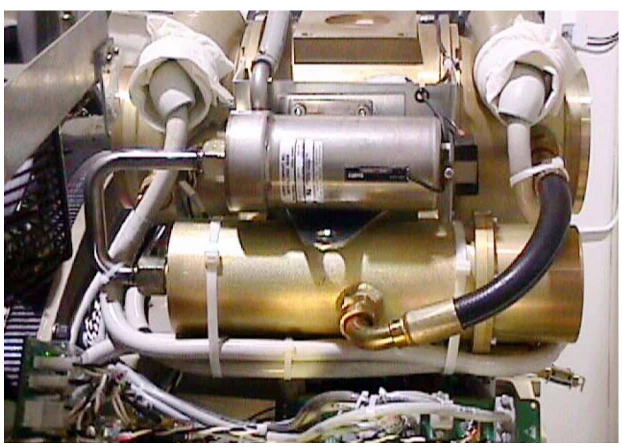


Figure 26 - HV Cables Properly Routed and Secured

14.) Align the cable terminal orienting key with the notch in the receptacle.
15.) Slowly insert the cable, to engage the connector pins, and seat the cable in the well.
- Tighten the cable locking ring.
- Rotate the cable strain relief for a clean cable dress.

NOTICE **Do not over tighten the locking ring. Over tightening can deform the cable plug sealing surfaces, break the oil seal between receptacle and housing, twist the receptacle, and disrupt internal wiring.**

NOTICE **IF YOU GET OIL ON YOUR HANDS, WASH THEM NOW.**

16.) Carefully wipe up all excess oil.
17.) Secure HV Cables using Large Ty-Raps, as shown in Figure 26.
18.) Disconnect hoist from tube and boom.
19.) Remove the post and boom from the gantry. Reference Figure 17, on page 20.
20.) Check for oil leaks:
   - Wrap rags or paper towels around the cable horns, and tape them into place.
   - Manually rotate the tube to the 6 o'clock position.
   - Return the tube to the 3 o'clock position
   - Remove the toweling and wipe up all excess oil.
   - Wipe off the cable horns, locking rings, and strain reliefs with a rag dampened with alcohol.
   - Repeat with a dry rag.
   - Wrap the cable strain reliefs and locking rings with a single layer of absorbent paper tissue.
   - You can use two inch wide strips cut from a paper napkin.
   - Wrap the bottom edge of the paper around the top end of the cable horn, and tape it into place.
   - Extend the top edge of the paper over the top of the locking ring, and tape it to the plastic cable strain relief.
   - Remove paper after leak check.
21.) Install the right gantry front cover bracket. Reference Figure 26, on page 27.
22.) Restore system power at the main disconnect panel.
23.) Turn on gantry **120 VAC**, **HVDC POWER** and **AXIAL DRIVE ENABLE** at the **STC** backplane. Wait at least 10 minutes to warm up the filament.

4 Alignment and Calibration

4.1 Z-Align using Beam on Window Alignment (BOW)

The purpose of the Z-Align procedure is to move the X-Ray Tube to be in alignment with the Detector and Collimator. The Detector and Collimator were aligned with the previous Tube, and Z-Align will correctly place the new Tube in the Z-Direction.

Note: For the BOW tool to work correctly, the X-Ray Tube needs to be close to proper alignment with the Collimator and Detector. If the Tube is not close to where it needs to be, the BOW tool will give erroneous data back. A POR scan and corresponding Tube movement can correct this problem.

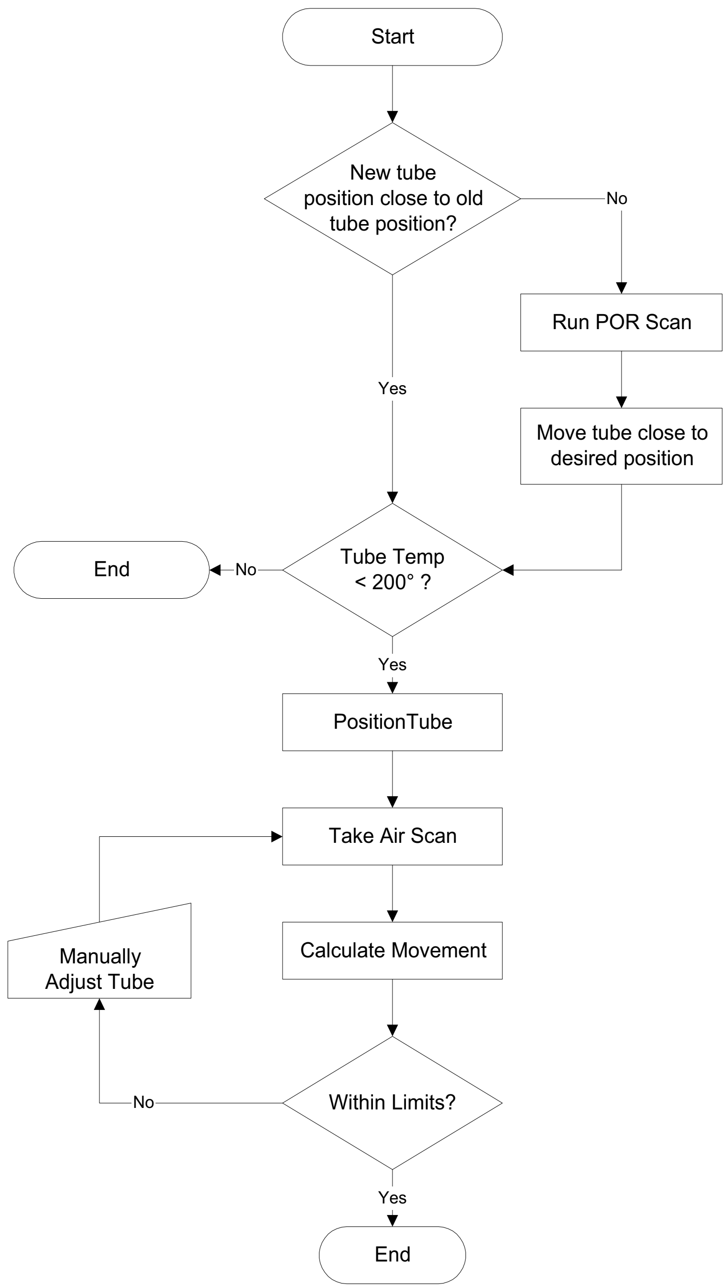


Figure 27 - BOW Alignment Process Flow

4.1.1 Verify Tube Temperature < 200° C

1.) Click DAILY PREP.
2.) Click TUBE WARM-UP.
3.) Click ACCEPT in pop-up window.
4.) Click PAUSE.
5.) Click CANCEL (do not QUIT Daily Prep).

NOTICE **Do not start X-ray exposures. Starting tube warm-up will result in additional tube cooling wait times.**

6.) Open system error log (`gesys_suite.log`).
7.) Click LAST PAGE.

Look for this type of entry:

```
This entry is created when Pause is clicked.
bay57 dailyPrepRx
StateMachineEventNotify.c 418
Tube temperature before Cold Tube Warmup is 315.15 degrees Celsius.
This entry is created when Cancel is clicked.
bay57 dailyPrepRx
StateMachineEventNotify.c 411
Tube temperature after Cold Tube Warmup is 312.89 degrees Celsius.
Start Tube Alignments when 200.00 degrees Celsius is reported.
bay57 dailyPrepRx
StateMachineEventNotify.c 418
Tube temperature before Cold Tube Warmup is 200.00 degrees Celsius.
```

8.) Repeat steps 3 through 5 as necessary to generate updates in the `gesyslog`.
9.) When tube temperature shows 200.00 Celsius, click QUIT to exit Daily Prep.

4.1.2 Accessing the Software

1.) Select SERVICE DESKTOP.
2.) Select REPLACEMENT
3.) Select TUBE CHANGE
4.) Select BOW ALIGNMENT

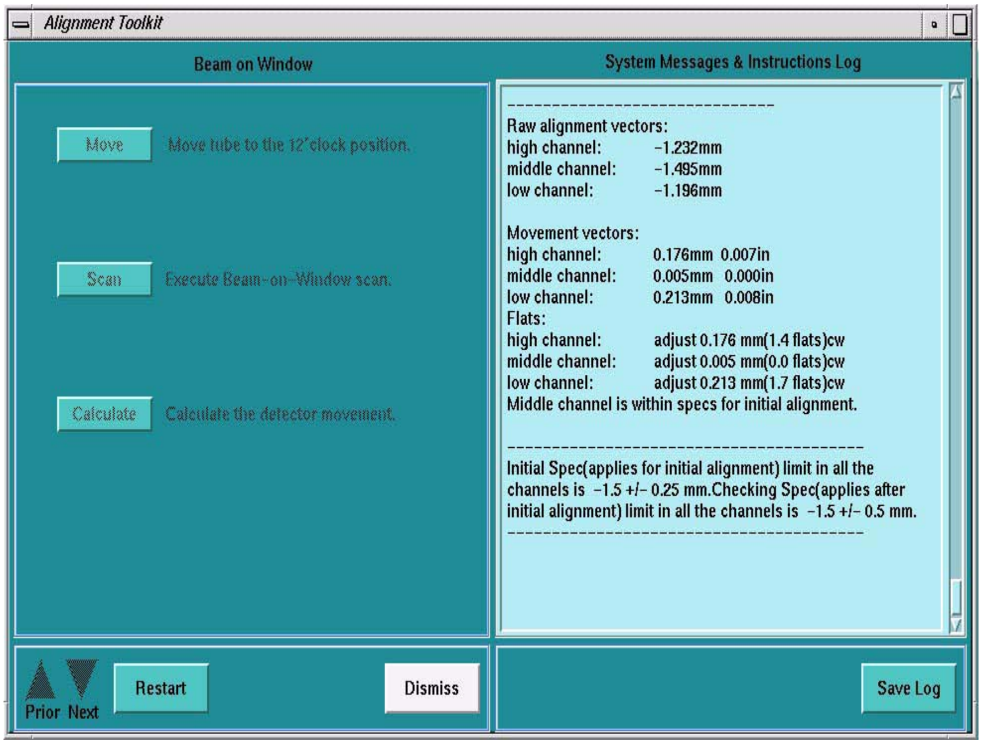


Figure 28 - Screen Shot for Beam on Window Alignment

4.1.3 Adjustment Procedure

1.) Place a dial gauge in the POR Dial Mount location.
2.) Select MOVE to send the tube to the 12 o'clock position.
3.) Acquire a Beam on Window Scan.
4.) Select CALCULATE.
5.) If adjustments are required, calculate the Tube Z-Movement to bring BOW into spec.

BOW Movement vectors:

- BOW high channel: -0.446mm -0.018in
- **middle channel: -0.653mm -0.026in**
- low channel: -0.611mm -0.024in

6.) After Selecting the CALCULATE button from the BOW Window, the GUI will display the `BOW Raw Alignment Vectors`, `BOW Movement Vectors` and the `Flats Movement Vector`.
7.) If the middle channel flats adjustment is more than 10 (either direction), run one POR adjustment to get the tube close. Then, proceed with Z-align, from the beginning. POR adjustment directions can be found in the applicable *System Service Manual*.
8.) Tube Z-Movement Value = BOW Middle Channel Movement Vector divided by five.
9.) Tube Z-Movement Direction:

a.) If the BOW Movement Channel value is Positive (+) move the Tube towards the REAR of the Gantry. Turn the POR nut CCW.

b.) If the BOW Movement Channel value is Negative (-) move the Tube towards the TABLE. Turn the POR nut CW.

Example: From the sample data above, the BOW Movement Vector for the Middle Channel is **-0.653mm or -0.026in**.

- Tube Z-Movement Value (mm) = -0.653 mm / 5 = -0.131 mm OR
- Tube Z-Movement Value (in) = -0.026 in / 5 = -0.0052 in.
- Tube Z-Movement Dir. negative = Towards the Table, turn the POR nut CW.

10.) Loosen the Tube Mounting Bolts.

11.) Move the Tube the desired Direction and Distance. Be as accurate as possible to avoid rescans.

12.) Tighten the Tube Mounting Bolts (49 ft lb)

13.) Select MOVE to send the tube to the 12 o'clock position.

14.) Acquire a Beam on Window Scan.

15.) Select CALCULATE

16.) If the Values are within spec, move on. If not, return to step 7.

Note: The specs for **BOW** are checked by the software. If an adjustment is needed for the first **BOW** scan, make adjustments per procedure. In the verification scan, after ISO alignments, the spec is different than the software version because the tube is warm. The spec is -1.5 ± 0.5mm.

4.2 Cold ISO Alignment

Cold ISO Alignment is done before the tube is heated by Generator calibrations. Hot ISO is done later. For details on Hot ISO, see Figure 4.4, on page 44.

Note: Use two dial gauges for ISO adjustment, to ensure POR accuracy while the tube is loose. This saves valuable time when calibrating for good image quality.

4.2.1 Verify Tube Temperature < 200° C

For details on how to verify Tube Temperature, see the procedure in section 4.1.1.

4.2.2 Accessing the Software

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT

3.) Select TUBE CHANGE

4.) Select ISO ALIGNMENT

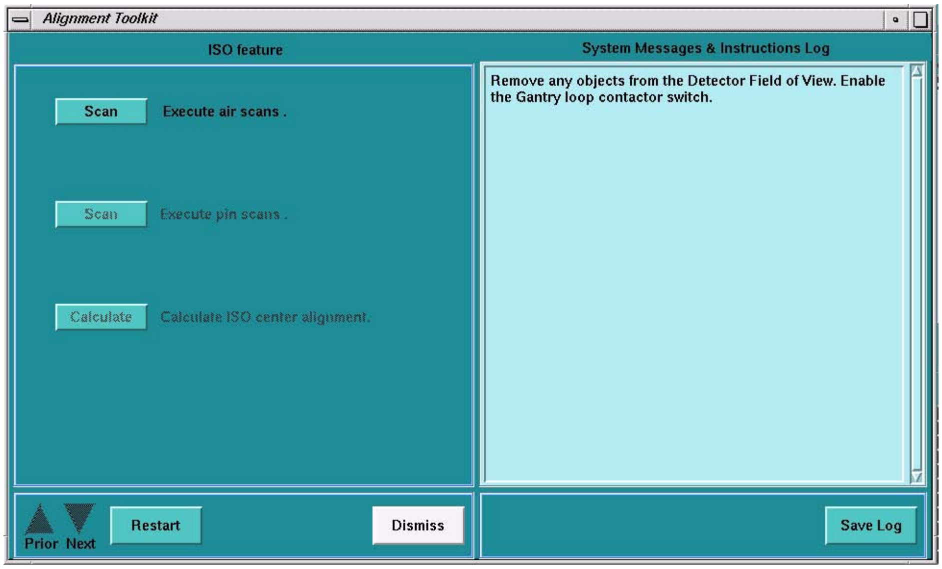


Figure 29 - ISO Alignment Menu

4.2.3 ISO Adjustment Procedure

1.) Execute Air scan (small spot).
2.) Execute Air scan (large spot).
3.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
4.) Execute pin scan (small spot).
5.) Execute pin scan (large spot).
6.) **CALCULATE** ISO center alignment.
7.) Mount two dial indicators on the tube assembly. (Figure 30.) Make sure that you zero them.

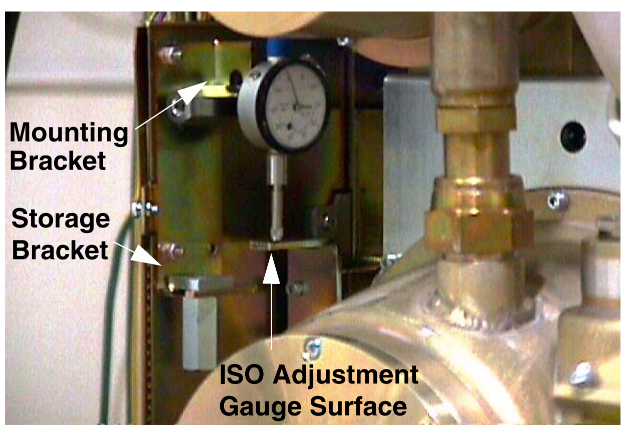


Figure 30 - ISO Dial Gauge Mounting Location

8.) Loosen the 4 M-12 bolts on the tube assembly a half turn.

9.) Adjust the tube UP / DOWN as indicated by the calculation. The adjustment bolt is located on the top of the tube - Please see Figure 31.

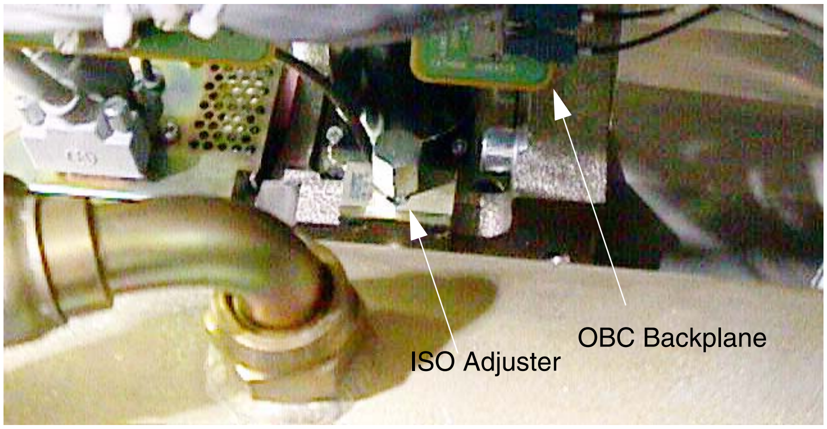


Figure 31 - ISO Alignment Adjuster

10.) Tighten the four (4) M12 bolts and verify ISO dial gauge still reads the correct adjustment value, and POR Gauge indicates no POR movement.
11.) Repeat steps 1 through 6.
12.) If the adjustments are within limit proceed to the next step, otherwise go to step 9.
13.) Tighten the four (4) M12 bolts. Torque to 49 lb-ft (66.4 Nm).
14.) If ISO adjustments were made, recheck BOW to verify that the Beam-on-Window values are within the drift spec of -1.5mm ±0.5mm.
- If BOW is within spec, DO NOT adjust anything
- If BOW is out of spec, Z-align, ISO and BOW recheck will need to be done.

4.3 Calibration - High Voltage

4.3.1 Access HV Maintenance through Service Desktop

1.) Select SERVICE DESKTOP.
2.) Select REPLACEMENT.
3.) Select HV SUBSYSTEM CHANGE.
4.) Select GENERATOR CHARACTERIZATION.

4.3.2 Generator Characterization

Use the Generator Characterization Program to update the “small spot” and “large spot” characterization files, to provide a starting point for the closed loop mode of the generator. This iterative process requires several scans at a different KV/MA/spot size. It calculates corrections, repeats the scan until the results fall within tolerance, then updates the characterization file.

Figure 32 - Proprietary Generator Characterization Menu Screen

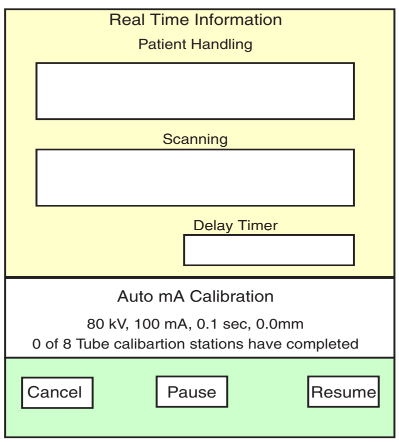


Figure 33 - Auto mA Calibration Status Screen

4.3.3 Verify kV Meter

This section describes the calibration check of system internal kV metering circuits.

1.) Select READ METERING.

2.) Select RUN to start the test. During the test, the firmware reads the metering circuits in the OFF state, then reads the metering circuits in the ON state, and finally reports the readings to the display.

3.) Compare the data in the “Delta” column on the Read Meter screen (Figure 9-4) to the data in the “Limit” column.

Note: “Delta” = DVM - A/D

CIRCUIT "OFF"	CIRCUIT "ON"
Anode kV = 0 ± 0.5	Anode kV = 50 ± 7.5
Cathode kV = 0 ± 0.5	Cathode kV = 50 ± 7.5
Total kV = 0 ± 0.5	Total kV = 100 ± 15.0

Table 2 - Generator Characterization Test Specifications

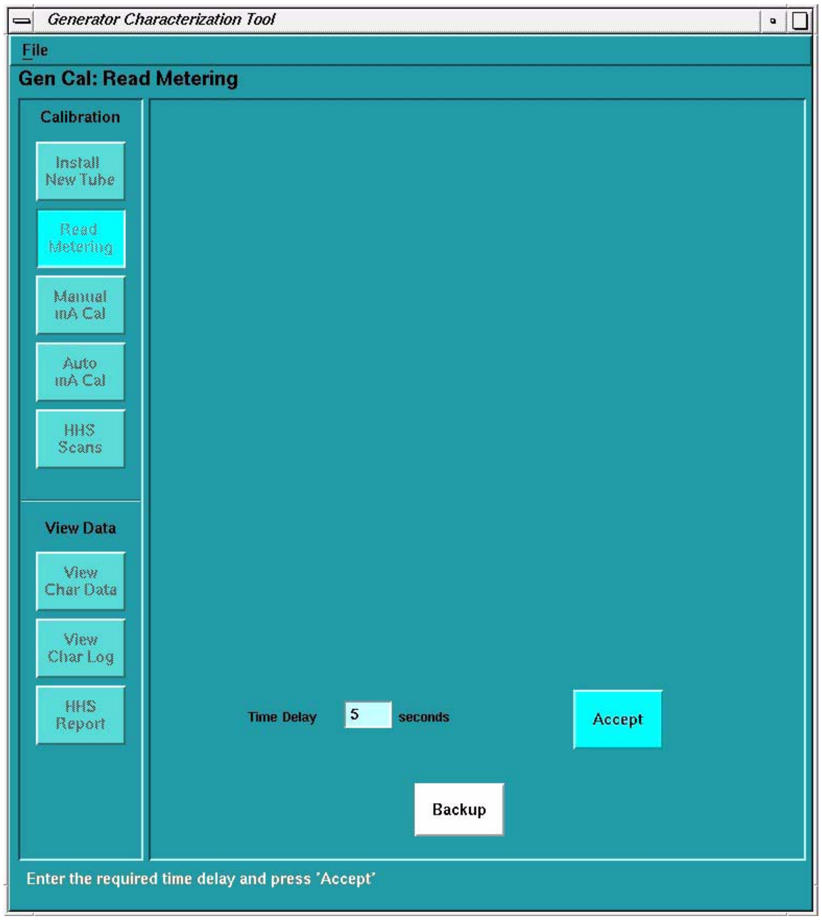


Figure 34 - Proprietary Read Metering Screen

4.3.4 Verify mA Meter – HHS Requirement

WARNING **WHEN SERVICING THE GANTRY:**

- **NEVER PUT ANY BODY PART INTO THE GANTRY WITHOUT FIRST DISABLING THE AXIAL DRIVE AND RE-VERIFYING (CHECK TWICE) THAT IT IS DISABLED.**
- **ENSURE THAT THE DRIVE STATUS LED (Figure 35) IS NOT LIT. DO NOT SERVICE THE GANTRY IF THIS LED IS ON.**

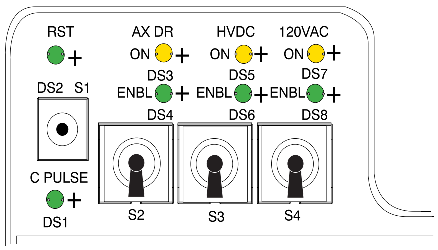


Figure 35 - STC Backplane Service Switches

This section describes the calibration check system internal mA metering circuits.

1.) Inside the Gantry:
   a.) Switch OFF the **HVDC ENABLE** on **STC** backplane.
   b.) Switch OFF the **AXIAL DRIVE ENABLE** on **STC** backplane.
   c.) Rotate the Anode tank to the 2 o'clock position.
   d.) Engage the gantry rotational lock.
2.) Select SERVICE DESKTOP.
3.) Select REPLACEMENT.
4.) Select HV SUBSYSTEM CHANGE.
5.) Select GENERATOR CHARACTERIZATION.
6.) Select READ METERING.

Note: On the display, enter a time delay in seconds, to provide enough time to walk from the console to the DVM, and record the reading. The test will not begin until this time delay expires.

Once it begins, the test enables the meter circuit for only 4 seconds.

7.) Use a DVM as an mA meter; connect it to the hardware on the anode side:
   a.) Connect the black lead to TP8 (ACAL1) on the mA board.
   b.) Connect the red lead to TP11 (ACAL2) on the mA board.

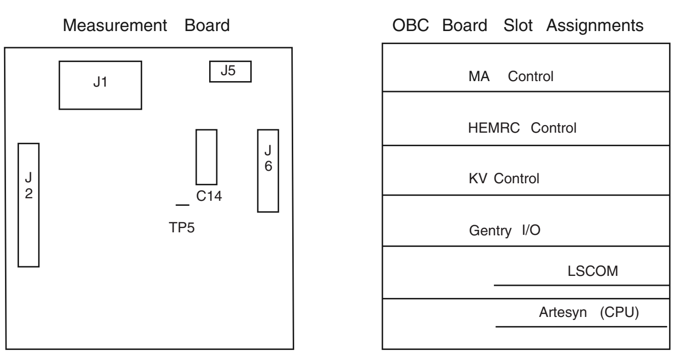


Figure 36 - Tank Measurement Board

8.) On the Display, select the ACCEPT button.

9.) Record the displayed, and measured, Anode mA values for “Circuit OFF” and “Circuit ON”.

Note: Your system has the test wire to TP5 included in the harness, the Cathode side should read approximately 19 mA during “Circuit On”.

10.) Disconnect the test equipment from the Anode side, if used.

Note: When you exit Generator Characterization, this test may generate `kV board tube spit counter = x error messages.`

11.) Use a DVM as an mA meter:

a.) Connect the black lead to TP9 (CCAL1) on the mA board.

b.) Connect the red lead to TP14 (CCAL2) on the mA board.

12.) On the Display, Select the ACCEPT button.

13.) Record the displayed, and measured, Cathode mA values for “Circuit OFF” and “Circuit ON”.

Note: Your system has the test wire to TP5 included in the harness, the Anode side should read approximately 20mA during “Circuit On”.

14.) Disconnect the test equipment from the Cathode side if used.

Note: When you exit Generator Characterization, this test may generate `kV board tube spit counter = x error messages.`

4.3.5 KV Gain Pots Adjustment – HHS Requirement

Install HV Divider

1.) Inside the Gantry on the **STC** backplane:

a.) Switch OFF the **HVDC ENABLE**.

b.) Switch OFF the **AXIAL DRIVE ENABLE**.

c.) Rotate the Tube to the 3 o’clock position

d.) Engage the gantry rotational lock.

e.) Switch OFF the **120 VAC GANTRY POWER**.

f.) Install the **HV** Divider between Tube and Tanks.

Note: Place the **HV** Divider on a table or tube hoist, so the cables reach the tube.

2.) Add a ground wire (minimum size of AWG 12) from Tube ground to bleeder ground. Refer to Figure 37.

CAUTION **Potential Electrical Hazard. Performix tube unit MUST be grounded to the gantry during testing.**

3.) Switch ON the **120 VAC GANTRY POWER**.

4.) Switch ON the **HVDC ENABLE**.

5.) Press the  button, on the gantry control panel, turning ON the drives power.

Note: If the gantry covers are removed press the **RESET BUTTON** on the **STC** backplane to turn ON Drives power.

6.) Reset the hardware.

NOTICE
Potential for tube damage

Incorrect installation of anode and cathode HV cables can destroy the Performix tube unit.

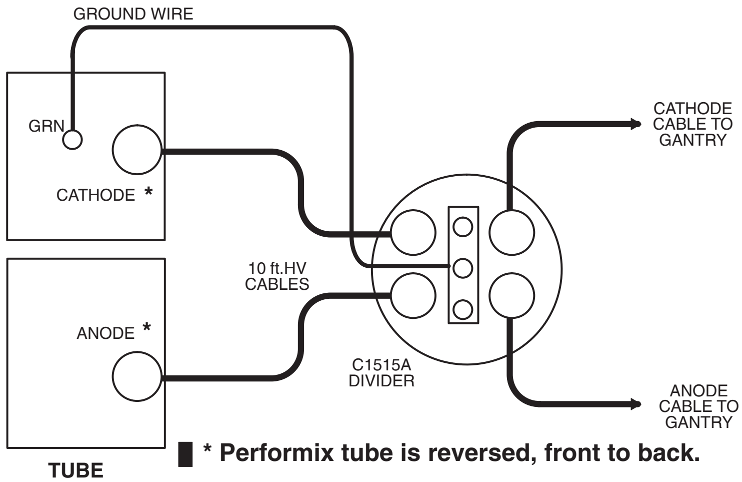


Figure 37 - HV Divider Installation

Set Up Instrumentation

Use an oscilloscope with 10X probes

1.) Use the Gantry Service Outlet to provide 120 Vac power for the scope. This will reduce noise on the scope waveform.

2.) Connect **channel one** to the **anode** output of the divider. Connect the scope ground to bleeder ground.

3.) Connect **channel two** to the **cathode** output of the divider. Connect the scope ground to bleeder ground.

Note: In order to minimize bleeder-induced ripple on the kV waveform, connect a 30 foot Belden shielded twisted pair cable between the scope probes and the bleeder.

4.) Trigger channel one, positive, DC couple, trigger mode normal.

5.) Channel one and two, 10v/div, time base 200ms.

6.) Invert Channel two.

Calibrate the Cathode

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT.

3.) Select HV SUBSYSTEM CHANGE.

4.) Select BLEEDER SETUP (DDC)

5.) Verify/Set-up the following **DDC** parameters:

- STATIC X-RAY ON	Interscan Delay **2.00**
- **1 SECOND**	**DAS** Gain **31**
- **1 SCAN**	Gantry Tilt **0.0**
- FOCAL SPOT LARGE	Trigger Rate 984
- **100 KV**	Filter BLOCKED
- **50 MA**	Calibration Vector NONE
- MONITOR ENABLE	

6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV equals 50 ± 0.5 kV.

Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.

7.) Adjust the Cathode pot on the kV board, until the scope reading for the Cathode kV, and the displayed reading for the Cathode kV in the message log, fall within ±0.5kV of each other.

8.) Use the pot, labeled CAKV, R316, on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
- CW increases the scope kV.
- 1/2 turn equals approximately 0.5 kV.

9.) Record the results on **FORM 4879**.

Calibrate the Anode

1.) Select SERVICE DESKTOP.
2.) Select REPLACEMENT.
3.) Select HV SUBSYSTEM CHANGE.
4.) Select BLEEDER SETUP (DDC)
5.) Verify/Set-up the following **DDC** parameters:

- STATIC X-RAY ON	Interscan Delay **2.00**
- **1 SECOND**	**DAS** Gain **31**
- **1 SCAN**	Gantry Tilt **0.0**
- FOCAL SPOT LARGE	Trigger Rate 984
- **100 KV**	Filter BLOCKED
- **50 MA**	Calibration Vector NONE
- MONITOR ENABLE	

6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV is 50 ± 0.5 kV.

Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.

7.) Adjust the Anode pot on the kV board, until the scope reading for the Anode kV, and the displayed reading for the Anode kV in the message log, fall within ±0.5kV of each other.

8.) Use the pot, labeled ANKV, R318, on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
- CW increases the scope kV
- 1/2 turn is approximately 0.5 kV.

9.) Record the results on **FORM 4879**.

Measure Total kV

1.) Select SERVICE DESKTOP.
2.) Select REPLACEMENT.
3.) Select HV SUBSYSTEM CHANGE.
4.) Select BLEEDER SETUP (DDC).

5.) Verify/Set-up the following **DDC** parameters:

- STATIC X-RAY ON	Interscan Delay **2.00**
- **1 SECOND**	**DAS** Gain **31**
- **1 SCAN**	Gantry Tilt **0.0**
- FOCAL SPOT LARGE	Trigger Rate 984
- **100 KV**	Filter BLOCKED
- **50 MA**	Calibration Vector NONE
- MONITOR ENABLE	

6.) Change the oscilloscope to add ch.1 and ch.2, to read total kV from the **HV** divider.

7.) Channel one and two, 20v/div, time base 200ms, trigger channel. one, positive.

8.) Select ACCEPT RX.

9.) Record the scope reading, and the Average. kV displayed in the message log, in **FORM 4879**.

10.) Display the Generator Characterization menu.

11.) Toggle the soft-key MONITOR ENABLE **OFF**, so the message log no longer displays kV and mA readings.

Verify kV Meter

Use this procedure to verify the calibration of the internal kV metering circuits.

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT.

3.) Select HV SUBSYSTEM CHANGE.

4.) Select GENERATOR CHARACTERIZATION.

5.) Select READ METERING.

6.) Select ACCEPT.

- The test begins after the time delay expires.
- Once the test begins, the software enables the meter circuit for 4 seconds.

7.) Record the displayed Anode kV, Cathode kV and Total kV values in the **FORM 4879** “Circuit OFF” and “Circuit ON” table.

8.) Select BACKUP.

Remove the External HV Divider

1.) Press the button, on the gantry control panel, turning **OFF** the drives power.

Note: If the gantry covers are removed, proceed to step 2.

2.) Inside the Gantry on the **STC** backplane:

a.) Switch OFF the **HVDC ENABLE**.

b.) Switch OFF the **AXIAL DRIVE ENABLE**.

c.) Switch OFF the **120 VAC GANTRY POWER**.

d.) Remove the **HV** Divider between the Tube and Tanks (Figure 37, on page 39).

e.) Reconnect the **HV** cables for normal operation.

NOTICE **Incorrect installation of anode and cathode HV cables can destroy the Performix tube.**

f.) Re-apply paper toweling around tube locking ring to absorb excess oil.

g.) Disengage the gantry rotational lock.

a.) Switch ON the **HVDC ENABLE**.

b.) Switch ON the **AXIAL DRIVE ENABLE**.

c.) Switch ON the **120 VAC GANTRY POWER**.

3.) Press the  button, on the gantry control panel, turning ON the drives power.

Note: If the gantry covers are removed, press the **RESET BUTTON** on the **STC** backplane to turn ON Drives power.

4.) Reset the hardware.

Install New Tube Program – HHS Requirement

Use this program to complete Auto mA Cal on a new tube. *Run this program only on a new tube.*

Refer to Figure 9-5.

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT.

3.) Select HV SUBSYSTEM CHANGE.

4.) Select GENERATOR CHARACTERIZATION.

5.) Select the INSTALL NEW TUBE.

Note: The system automatically warms up the tube.

6.) The system prompts you with the tube type. Verify the number corresponds to your tube type; answer `Y` or `N`.

SOFTWARE TOKEN	HOUSING NUMBER	INSERT NUMBER
12-MX_135CT	46-274800G1	46-274600G1
13-MX_165CT	46-309500G2	46-309300G1
14-MX_165CT_I	46-309500G2	46-309300G2
15-MX_200CT	2137130-2	2120785

Table 3 - Tube Type Table (SW tokens for various Housings and Inserts)

7.) Press **START SCAN** when it flashes, to automatically run the program and update the display:

`seed filament current shift scans`

Auto mA Calibration

Run this program when you replace the X-Ray tube, or the system requires re-calibration.

Note: If you have just completed running the Install New Tube Program in the previous section, you do NOT need to run Auto mA Cal.

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT.

3.) Select HV SUBSYSTEM CHANGE.

4.) Select GENERATOR CHARACTERIZATION.

5.) Select AUTO MA CAL.

Note: The software automatically warms up the tube.

6.) Press **START SCAN** when it flashes, to automatically run the program and update the display:

– `Ductility warm-up -`

– `Auto mA Cal -`

7.) The system displays the final filament currents on the screen.

---

kV AND mA Verification Data Sheet – HHS Requirement

Access the software and perform the following:

1.) Select SERVICE DESKTOP.

2.) Select REPLACEMENT.

3.) Select TUBE CHANGE.

4.) Select HHS SCANS.

When Scans are complete:

5.) Select HHS REPORT.

Insure all kV and mA stations meet specifications found in form 4879. (Form can be found in the appropriate system Advanced Service manual.)

Note: If your system has InSite connectivity, you can have a copy of the HHS report e-mailed to your laptop for printing by sending a sweep report to the autoSC.

Send an e-mail in the following format:

TO: autosc (MED) OR autosc@olc.med.ge.com

SUBJECT: sweep

Message content:

type = get_hhs_report

system id = xxxxx

4.4 HOT ISO Alignment

4.4.1 Accessing the Software

1.) Select SERVICE DESKTOP.
2.) Select CALIBRATION.
3.) Select HOT ISO ALIGNMENT.

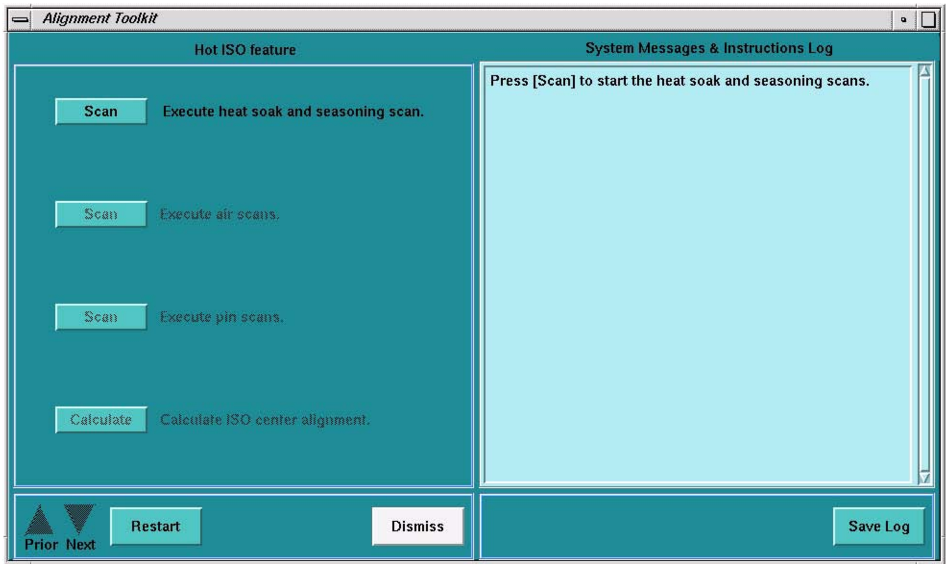


Figure 38 - HOT ISO Screen

4.4.2 Adjustment Procedure

1.) Execute the Heat Soak and Seasoning scan.
2.) Execute Air Scan (small spot).
3.) Execute Air Scan (large spot).
4.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
5.) Execute Pin Scan (small spot).
6.) Execute Pin Scan (Large spot)
7.) Calculate correction factor with Calculate button.

Note: No movement of the tube is required. This is a software correction done automatically.

4.5 New Tube Configure

Access the software and perform the following:

1.) Select SERVICE DESKTOP.
2.) Select REPLACEMENT.
3.) Select TUBE CHANGE.
4.) Select NEW TUBE CONFIG.
5.) Select SCAN HARDWARE RESET and reset application SCAN.

4.6 DAS Gain Calibration

1.) Perform **DAS** Gain Calibration by selecting the function from Scanner Utilities (**left head**).
2.) Select SCANNER UTILITIES, and Select DAS GAIN CALIBRATION**.**
3.) Ensure that there is nothing in the x-ray beam and continue.
4.) The system will now perform a Mylar window check and provide the appropriate messages if the window should need cleaning.
5.) Upon completion of the Mylar window scans the system will now take 31 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

4.7 Collimator Calibration

1.) Perform Collimator Calibration by selecting the function from Scanner Utilities (**left head**).
2.) Select SCANNER UTILITIES, and Select COLLIMATOR CALIBRATION.
3.) Ensure that there is nothing in the x-ray beam and continue.
4.) The system will now perform a Mylar window check if needed and provide the appropriate messages if the window should need cleaning.
5.) Upon completion of the Mylar window scans the system will now take 8 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

4.8 Detector Z-Slope Calibration

1.) Perform Z-Slope Calibration by selecting the function from Scanner Utilities.
2.) Select SCANNER UTILITIES, and select Z-SLOPE CALIBRATION.
3.) Ensure there is nothing in the x-ray beam and continue.
4.) The system will now scan and provide the appropriate messages if the calibration should fail.

4.9 Calibration Process

CAUTION Potential for IQ Artifacts

If kV Gain Pots were adjusted in section 4.3.5, you cannot use Fastcal. A full set of calibrations must be performed.

1.) Perform a Gantry Balance check.
   a.) SERVICE DESKTOP > REPLACEMENT > GANTRY SERVICE BALANCE
   b.) Check that the tube counterbalance weights are installed and bolts are secure.
   c.) Complete the Gantry balance check.
2.) If this test does not pass, it is not recommended that you perform calibrations until the balance issue is resolved. Refer to the balance procedure in the Gantry Section of the appropriate Advanced Service Manual.
3.) Bring up the main menu: It is represented as an icon located on the bottom of the screen labeled as “Daily Prep”. Click the on-screen DAILY PREP button (**left head**).

CAUTION Potential for IQ Artifacts

If Fastcal has been run in the past 48 hours, only one filament will be selected. In this case, Fastcal will need to be run twice in order to calibrate both filament sections. It is suggested that you watch the Fastcal scan list as scans are being taken to insure that both filaments are being scanned.

Note: To check when the last Fastcal was run:

a.) Open a unix shell.

b.) `cd /usr/g/service/state`

c.) `tail -50 ssw.calreport.hist`

4.) Select and run FASTCALIBRATION.

5.) The system will run through the Normal Fast Calibration Procedure.

4.10 CT N# Adjust

1.) After the Fast Calibration is run, run CT N# Adjust Scans.

2.) Select SCANNER UTILITIES, and select AUTO CT NUMBER ADJUST.

3.) Place the 20 cm water phantom on the phantom holder and center the phantom.

4.) Run the CT N# Adjust scans.

CAUTION Potential for IQ artifacts

Because DETAILED CALIBRATIONS were not run, it is IMPERATIVE to visually inspect all CT N# Adjust Images for artifacts.

5.) After all Scans have completed, visually inspect all images for artifacts. If artifacts are seen, take note of the scan configuration, i.e. 4 x 2.5 / 200 mA / 120 kV / 2s

4.11 Quick Tube Change IQ Scans and Image Analysis

4.11.1 Quick Tube Change IQ Scans – Large FOV

Purpose: Because Auto CT Number Adjust scans only look at the small field of view, there needs to be a series of scans taken with the large field of view. The 35cm and 48cm Phantoms will be used. Here is a listing of scans, and the procedure to run Image Analysis.

TEST	PROTOCOL	SERIES DESCRIPTION	4-SLICE	8-SLICE	16-SLICE
QA3 - 120kV - 10mm - 2i	Image Series QA	QA#3/Small/1.0s	X		
QA3 - 120kV - 8x2.50		QA#3/Small/8x2.5		X	
QA3 - 120kV - 16x0.625		QA#3/Small/16x0.625			X
QA3 - thintwin or 0.625/2i		QA#3/Small/2x0.625/2s	X	X	X
P35 - 4x1.25	Image Series 35cm Poly	35cm/4x1.25/2s	X		
P35 - 8x1.25		35cm/8x1.25/2s		X	
P35 - 16x1.25		35cm/16x1.25/2s			X
P48 - 4x2.5	Image Series 48cm Poly	48cm/4x2.5mm/1s	X		
P48 - 8x2.5/120/200/1s		48cm/8x2.5mm/1s		X	X

Table 4 - Quick Tube Change IQ Scans

1.) The phantom under test must be centered before starting scanning.

2.) Select SCANNER UTILITIES and CENTER PHANTOM.

3.) Repeat using the Center Phantom function until the phantom is centered.

4.) Exit the Center Phantom Utility and The Scanner Utilities Screen.

5.) Select NEW PATIENT.
6.) Enter a PATIENT ID, and PATIENT NAME.
7.) Select the SERVICE tab from the Anatomical Selector.
8.) Select MANUFACTURING.
9.) Select the Protocol being investigated (Image Series QA, Image Series 35 cm Poly or Image Series 48 cm).
10.) Hit the NEXT SERIES button until the desired system configuration appears.
11.) Scan the series and move on to the next desired system configuration.
12.) When the IQ scans are done for a phantom, select END EXAM.
13.) Return to Step 1 and repeat for all phantoms being scanned.
14.) When all Phantoms have been scanned from the Service Desktop, select IMAGE QUALITY and IMAGE ANALYSIS.
15.) Select the small AUTO 1X button at the top of the screen.
16.) Select the pull down menu AUTO 1X, select the Phantom to Analyze (ImgSer 48cm or ImgSer 35cm), and select the Scan Configuration to Analyze, and select the AUTO button.
17.) In the SORT WINDOW, make sure the exam being Analyzed is highlighted.
18.) Back in the Image Analysis Window click on the ACCEPT button.
19.) A PASS/FAIL window will appear, along with details of what Passed and Failed.
20.) If the scan passed, that system configuration is in an acceptable state. Repeat steps 16 through 19 until all scans have been analyzed.
21.) If any scans fail, take note of the system configuration.

4.11.2 Image Quality Check for Form 4879 - HHS Requirement

In addition to the Quick Tube Change IQ Scans, you will also need to refer to the FORM 4879 that is appropriate for your scanner and scan the phantoms required at the prescribed techniques in order to complete FORM 4879.

4.12 For Artifacts and Image Analysis Failures

Purpose: If there were no Artifacts or Image Analysis Failures, the Quick Tube Replacement Procedure is complete. Follow the following instructions if you have Artifacts or Failures.

1.) Compile your notes from CT N Adjust Artifacts and Image Analysis Failures.
2.) Select SCANNER UTILITIES and DETAILED CALIBRATION.
3.) A Gantry Balance Check will run.
4.) From the Detailed Calibration list, select only the scan configurations that failed.

Example: There was a CT Number Adjust Artifact at 120kV with a 4x1.25 scan configuration, and there was an Image Analysis Failure at 120kV at 4x2.5, the 120KV button and the 4x1.25 and 4x2.5 slice collimation buttons would be selected. Leave all of the SFOV and FOCAL SPOT buttons selected.

5.) Run the Detailed Calibrations.
6.) Once Detailed Calibrations are done, re-run AUTO CT NUMBER ADJUST. Run the same system configurations that were run for Detailed Calibrations.
7.) Check to ensure that the CT Number Adjust scans are free of artifacts and make sure that Image Analysis now passes on the previously failed configurations.

4.13 Finalization

Save system state.

5 Gantry Rotation Safety Check

Use DDC to Rotate the Gantry for 10 minutes, at the fastest allowed gantry speed.

1.) Launch Diagnostic Data Collection (DDC) Utility:

a.) Open the Common Service Desktop

b.) Click on the DIAGNOSITICS Icon

c.) Click on the DATA ACQUISITION Folder

d.) Click on the DIAGNOSTIC DATA COLLECTION Menu Item

2.) Prepare System for DDC Scan. Refer to Figure 39.

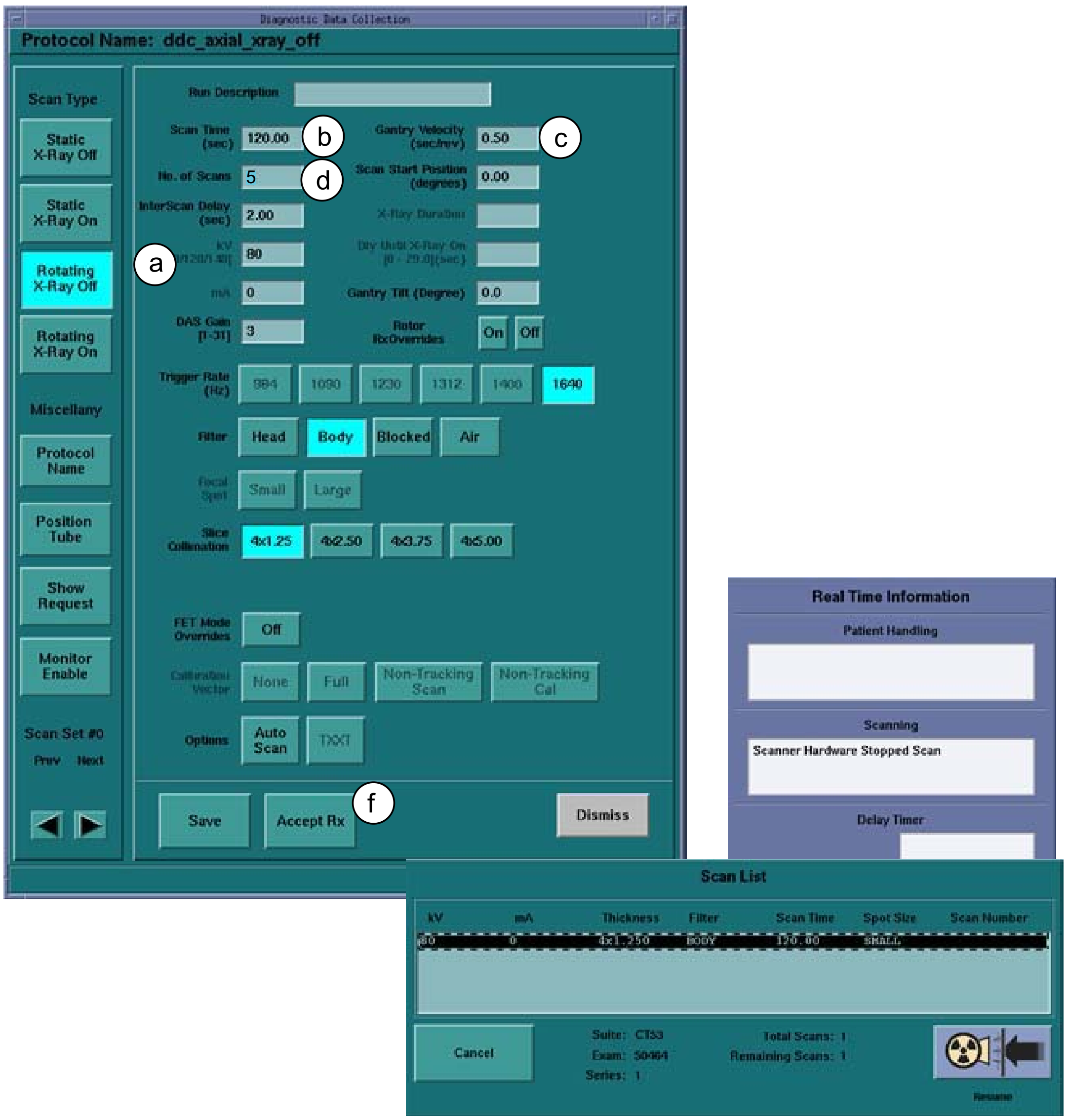


Figure 39 - DDC Tool

a.) Select ROTATING X-RAY OFF

b.) Type: `120` in `Scan Time (sec)` field

c.) Type `0` in the `Gantry Velocity (sec/rev)` field and Press **ENTER**. The system will then default to the fastest allowable service mode revolution time.

d.) `No of Scans` = **5**. This will result in 10 minutes of rotation.

e.) Verify other settings, as shown.

f.) Click on ACCEPT RX button, to send scan request.

The Utility:

- Displays the Scan List GUI and the Real Time Information window.
- After a few seconds, it starts flashing the **SCAN** button on the console.

3.) Press the Scan button on the Console, to start each DDC Scan. When complete, the DDC Utility dismisses the Scan List GUI and the Real Time Information window.

4.) If no issues arise, gantry rotation capability is safe for use.

Appendix B - Global Installed Base (GIB) Update Instructions

1 United States Instructions

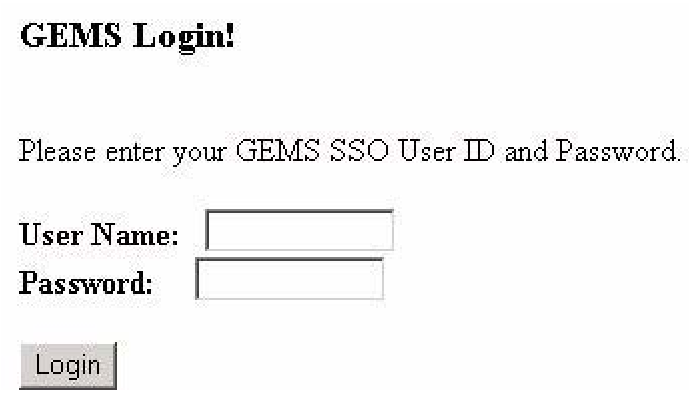
NOTICE **The instructions in this section are for the Americas Pole only.**
Instructions for the Europe Pole may be found in Section 2 (on page 54).
Likewise, instructions for the Asia Pole may be found in Section 3 (on page 54).

These instructions take you through the steps required to login to the GIB system via the web and update an installed CT system for Tube Casing and Tube Insert Serial Numbers.

Use the link below to access the GIB web site.

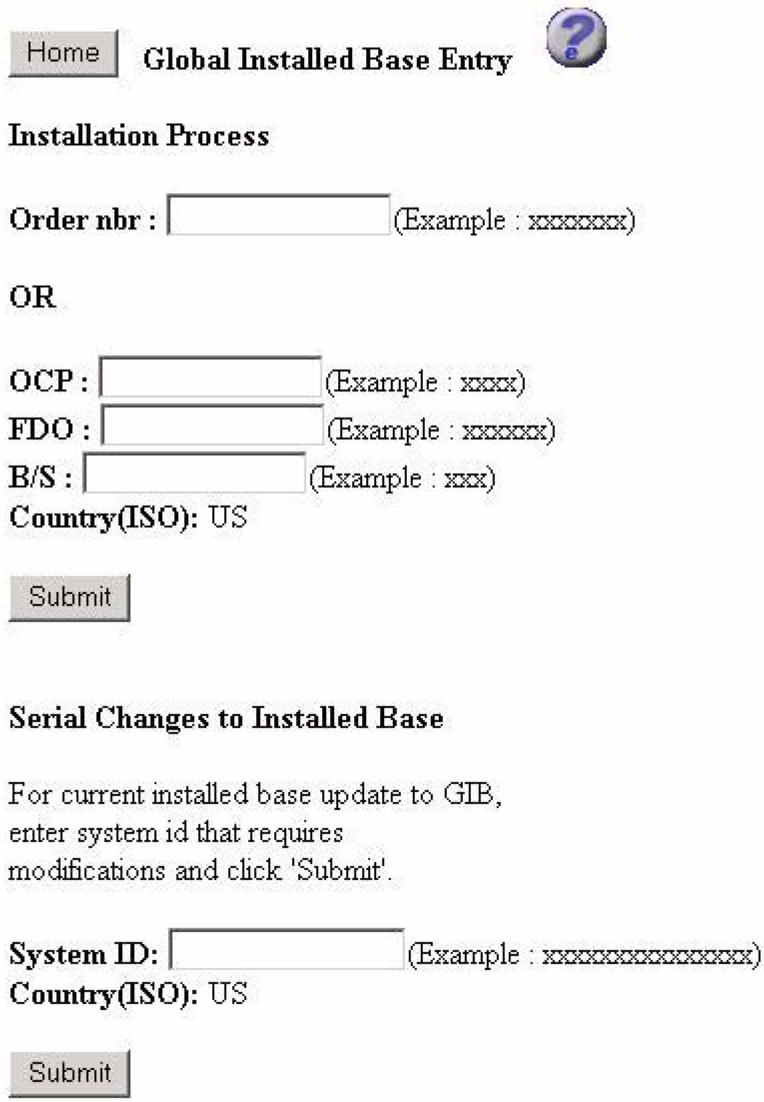
http://egems.gemedicalsystems.com/gib/gib_MainMenu.jsp

1.) The first screen you will see is the login screen. Log in using your SSO and password.

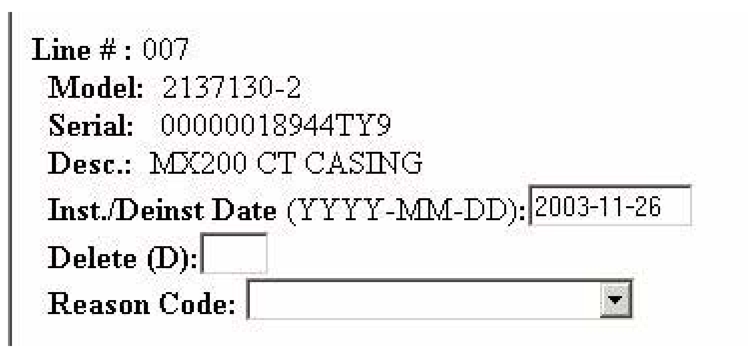


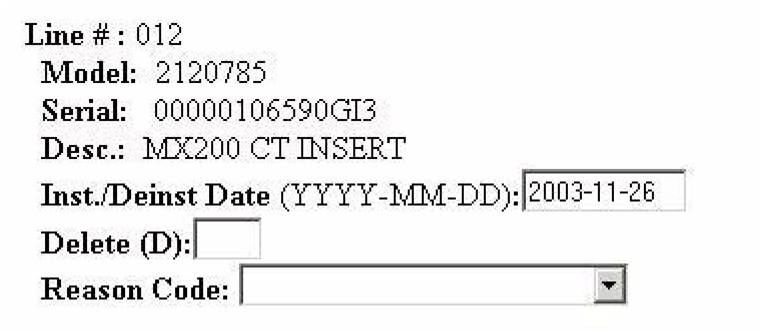
2.) After logging in, the next screen you will see is shown below. Select “Global Installed Base entries”.

3.) The next screen shown below is the Main entry screen. Enter your GE Cares System ID in the field provided at the bottom of the screen and select “Submit”.



4.) This will pull up a listing of Serial numbers attached to the CT system ID. Below is an example of the CT Casing and Insert Serial Numbers in the format you will see. The Line # shown is purely for list purposes and just depends on the order the serial numbers were originally entered for this system.





5.) To delete Serial numbers for Tube Casings and Inserts that no longer exist on this system, enter a “D” in the Delete field shown in the above examples. When you have done this for any items on the current displayed page, select “Submit” to update GIB.

6.) The screen below shows the “Submit” button and in this instance that there are 3 pages of serial numbers attached to this system.

2 Europe Instructions

For the system being inspected, document the following information and send it by e-mail to the address listed below.

- System country name:
- Gantry Model Number:
- Gantry Serial Number:
- Tube Case Model Number:
- Tube Case Serial Number:
- Tube Insert Model Number:
- Tube Insert Serial Number:
- MUST System ID:

Send this information to: ***GIB, European Central Admin (MED)***

Please note in the e-mail to have any other Tube or Insert entries not listed above to be removed from GIB for this system.

3 Canada/Latin America/Asia Instructions

Follow local procedure to update the Tube Case and Tube Insert model and serial numbers for the CT system being inspected.

Remove any Tube or Insert entries in GIB that do not match the tube currently on the system.